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REVIEW

Microspheres: Preparation Methods, Advances, Applications, and Challenges in Drug Delivery

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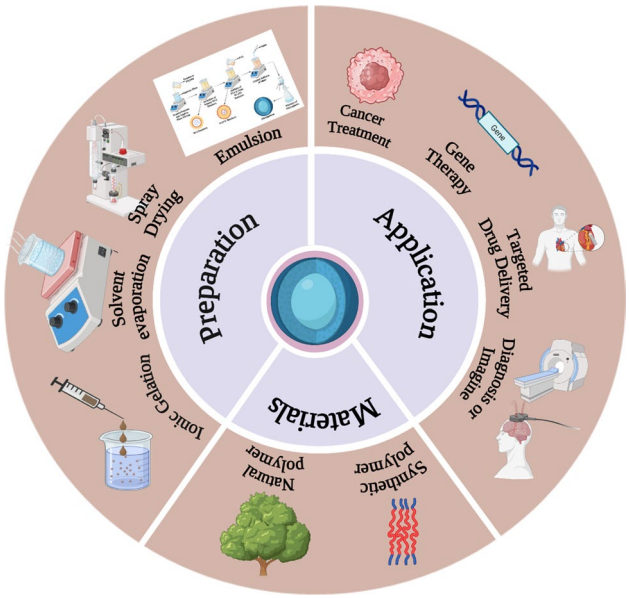
Abstract

Microspheres used as a versatile and efficient drug delivery system. These spherical, biodegradable particles, made from synthetic or natural polymers, range in size from 1 to 1000 µm and offer controlled and targeted drug release. They are widely used in fields such as drug delivery, bone tissue manufacturing, and environmental decontamination. Microspheres are categorized based on their properties and functions, including bio adhesive, magnetic, floating, radioactive, and polymeric types, each contributing to enhanced drug delivery by increasing bioavailability, reducing side effects, and improving patient compliance. Various preparation methods, such as single and double emulsion techniques, polymerization, and spray drying, play a crucial role in their efficiency, while evaluation parameters like particle size, drug entrapment efficiency, and in vitro release studies assess their performance. Recent advancements have significantly enhanced microsphere technology, including superhydrophobic surfaces (SHS) for improved fabrication in photonic devices, protein crystallization, and medical diagnostics. Probiotics-loaded microspheres have shown increased stability and viability in cosmetic and therapeutic formulations. In cancer therapy, microspheres are effectively used in embolization techniques, glioblastoma treatments, and targeted drug delivery systems for lung and prostate cancers. Additionally, eco-friendly dextran microbeads are being utilized in multiplex assays, while optofluidic immunochips aid in detecting myocardial infarction biomarkers. Despite these advancements, challenges such as ensuring stability, uniform size, material selection, and reproducibility remain significant hurdles in developing effective microsphere-based therapies and products.

Graphical Abstract



Extended author information available on the last page of the article

Keywords Microspheres · Metrics and encapsulation · Polymer · Emulsion crosslinking · Recent advancement

Introduction

The new drug delivery system aims to deliver the medication at rates that are adjusted to the body's needs during treatment, while also making sure that the therapeutic agent reaches its intended site of action as soon as possible. Because advanced drug delivery systems (DDS) allow precise control over drug release rates and can target specific bodily locations, they have had a significant impact on the healthcare industry. The best drug delivery device targets the active ingredient to the right location and delivers medication at a controlled pace that adapts to the body's needs. Drug carrier technology offers an advanced method of drug delivery by employing carrier particles like lipids, nanoparticles, or microspheres [1, 2]. Microspheres are defined as spherical particles ranging in size from 1 to 1000 µm, composed of natural or synthetic polymers. They are extensively utilized in pharmaceutical formulations due to their ability to provide controlled and targeted drug release. Their versatility allows them to encapsulate a wide range of therapeutic agents, ensuring stability, prolonged activity, and improved bioavailability. Microspheres play a crucial role in reducing systemic toxicity, enhancing therapeutic efficacy, and improving patient compliance, making them an indispensable component in modern drug delivery systems [3]. Two different kinds of drug delivery systems are represented by micrometrics and microcapsules. A uniform distribution of the drug within the material is made possible by the drug's dispersion throughout the microsphere's matrix in micrometrics. On the other hand, the drug is contained within a discrete capsule wall in microcapsules, which clearly separates the substance from its surroundings. One possible application for controlled drug release is in solid biodegradable microspheres, where the drug is dissolved or distributed throughout the matrix. They have a spherical, free-flowing shape and are made of synthetic polymers or biodegradable protein [3, 4]. Microspheres have demonstrated remarkable versatility in drug delivery systems, diagnostics, and regenerative medicine, significantly advancing targeted therapies. Recent statistics highlight their growing significance, with studies showing that microsphere-based delivery systems have increased therapeutic efficacy by up to 30% compared to conventional formulationslly, clinical case studies have validated their effectiveness in cancer therapies, such as glioblastoma and lung cancer, where microsphere-mediated treatments have shown enhanced localized drug release and reduced systemic toxicity. Microsphere widely used in regenerative medicine for tissue scaffolding and in diagnostics as imaging agents, contributing to improved patient outcomes [5].

Microspheres have emerged as a versatile drug delivery system due to their ability to encapsulate both hydrophilic and hydrophobic drugs, ensuring controlled and targeted release. 'Their significance lies in addressing critical challenges in modern therapeutics, such as enhancing bioavailability, reducing dosing frequency, and minimizing systemic side effects.' Additionally, the use of natural and synthetic polymers as sources for microsphere fabrication has opened new avenues for creating biocompatible and biodegradable delivery systems. A growing body of literature supports the utility of microspheres in diverse medical applications. Studies have demonstrated their potential in cancer therapy, where microspheres loaded with chemotherapeutic agents enable localized drug delivery, reducing toxicity to healthy tissues. Similarly, their application in regenerative medicine, such as scaffolds for tissue engineering, highlights their adaptability beyond traditional drug delivery. By exploring potential sources like bioadhesive polymers, magnetic materials, and radioactive isotopes, researchers have developed specialized microspheres for various therapeutic and diagnostic purposes [5]. The study of these systems is pivotal for advancing personalized medicine and improving patient outcomes in chronic and complex diseases.

In recent years, microspheres have gained significant attention for their role in advanced drug delivery systems, particularly in personalized medicine and targeted therapies. Biodegradable microspheres, such as those made from PLGA and chitosan, have demonstrated improved therapeutic outcomes, including prolonged drug release and reduced toxicity. Recent studies, such as the use of Yttrium-90 radioactive microspheres for liver cancer treatment, highlight their success in clinical applications. Novel preparation techniques like microfluidics and 3D printing have enabled precise control over microsphere size and drug release profiles, expanding their potential in cancer therapies, regenerative medicine, and vaccine delivery. Despite these advancements, challenges such as low encapsulation efficiency, scalability issues, and environmental concerns remain critical areas of focus [6]. Recent research has emphasized optimization strategies, such as enhancing double emulsion techniques and developing eco-friendly preparation methods, to address these limitations and pave the way for industrial-scale applications.

Advantages

Controlled Release

Microspheres can be designed to achieve sustained release, allowing medications to be gradually and consistently released over an extended period. Microspheres offer controlled drug delivery, improved patient compliance, reduced systemic side effects, and enhanced drug stability. Additionally, by modifying their surface characteristics, microspheres can be tailored for targeted delivery to specific tissues or cells. This targeted approach enhances the therapeutic efficacy of the medication by concentrating its effects at the desired site while simultaneously minimizing potential side effects in non-target areas. These combined features make microspheres a versatile and effective drug delivery system in modern medical applications [7, 8].

Enhanced Stability

Microspheres play a crucial role in protecting active ingredients from environmental factors such as light, oxygen, and moisture, which can degrade sensitive medications. By providing this protective barrier, microspheres help to extend the shelf life and maintain the stability of the medication, ensuring its effectiveness over time. Furthermore, they can significantly enhance the bioavailability of drugs that are poorly soluble in water. Encapsulation within microspheres improves the solubility of these drugs, allowing for better absorption in the body and increased therapeutic efficacy. These benefits make microspheres an invaluable tool in pharmaceutical formulations [9].

Minimization of Side Effects

Microspheres offer the advantage of localized drug delivery, enabling medications to be administered directly to the target site. This precision reduces systemic exposure, ensuring that the therapeutic effect is concentrated where it is needed most while minimizing side effects that often arise from non-targeted drug delivery. Additionally, by controlling both the release rate and the specific location of the drug, microspheres help to reduce the risk of toxicity. This controlled approach ensures that the drug is delivered in optimal amounts over a desired period, enhancing safety and effectiveness in treatment. These attributes highlight the potential of microspheres in improving the precision and tolerability of drug therapies [10].

Drug Delivery Systems

Microspheres are a highly versatile component of drug delivery systems, as they can be tailored for oral, intravenous, and topical administration, catering to a wide range of therapeutic needs. Their adaptability stems from their ability to encapsulate a variety of drugs, including hydrophilic, hydrophobic, and biologically sensitive molecules, ensuring stability and controlled release across different routes of administration. In oral drug delivery, microspheres enhance bioavailability and protect drugs from degradation in the gastrointestinal tract. For intravenous use, they allow precise control over drug release and targeted delivery to specific tissues or cells, reducing systemic side effects. In topical applications, microspheres facilitate sustained and localized release, improving treatment efficacy for skin conditions or localized infections. This flexibility, combined with their ability to enhance drug stability, bioavailability, and targeting, makes microspheres an essential tool in modern pharmaceutical innovation [11]. Microspheres offer the unique advantage of tailored drug release profiles, where the release rate can be precisely adjusted by altering their composition, structure, or polymer matrix. This flexibility enables the customization of drug delivery systems to meet specific therapeutic needs, such as rapid release for immediate relief or sustained release for prolonged treatment, reducing dosing frequency and enhancing patient compliance. Additionally, surface modifications on microspheres further expand their functionality by improving targeting capabilities, enhancing cellular uptake, and increasing stability within biological systems. These surface alterations, such as the addition of ligands or functional groups, enable microspheres to interact specifically with targeted tissues or cells, maximizing therapeutic effects while minimizing off-target side effects [12]. Together, these features make microspheres a powerful platform for precision medicine and personalized therapy.

Biocompatibility and Safety

Microspheres, particularly those crafted from biocompatible materials, are designed to exhibit reduced immunogenicity, making them well-tolerated within the human body. This low immunogenic response ensures that they can be used safely without triggering significant immune reactions, even in sensitive applications. Moreover, biodegradable microspheres provide additional safety and convenience for long-term use, as they naturally degrade into non-toxic byproducts such as carbon dioxide and water. This biodegradability eliminates the need for surgical removal and minimizes the risk of adverse reactions over extended periods. These characteristics make microspheres highly suitable for a range of medical applications, including sustained drug delivery, tissue engineering, and therapeutic interventions requiring prolonged activity or repeated administration [13, 14].

Enhanced Diagnostic Capabilities

Microspheres have advanced applications in imaging and tracing due to their ability to be labeled with imaging agents, enabling precise tracking in the body using techniques such as MRI and PET scans. Recent research has explored hybrid PET/MRI systems, which combine the high anatomical resolution of MRI with the functional imaging capabilities of PET. This synergy enhances diagnostic accuracy in conditions like cancer, where precise mapping of tumor boundaries and treatment response is critical. Innovations include the development of disease-specific tracers that improve targeting and reduce non-specific background signals. Recent advancements in materials science have led to the development of phosphate-based glass microspheres with tunable properties, offering significant potential in medical imaging and therapy. These microspheres, enriched with elements such as manganese or copper, serve as dual-purpose agents by enhancing both T1 and T2 MRI contrast, enabling more detailed and accurate imaging [6]. Furthermore, their porous structures provide opportunities for integrating imaging with drug delivery, allowing the creation of theranostic tools that combine diagnosis and treatment in a single system. This dual functionality not only improves diagnostic precision but also facilitates targeted therapy, underscoring the transformative potential of microspheres in medical imaging and personalized healthcare [6].

Reduced Drug Degradation

Encapsulating drugs in microspheres provides substantial advantages by protecting them from environmental factors such as light, oxygen, and moisture that can cause degradation. This protective shield ensures the drug's stability, preserving its efficacy and extending its shelf life until it reaches its target site. In addition, microspheres support combination therapy by enabling the concurrent delivery of multiple drugs, each targeting distinct disease pathways or mechanisms. This multi-drug delivery system not only enhances therapeutic outcomes but also simplifies treatment regimens, which in turn improves patient compliance. By offering a controlled release and allowing targeted drug action, microspheres significantly contribute to the development of personalized and effective treatment strategies, particularly in complex conditions where multiple interventions are required [15].

Disadvantages

Complex Manufacturing Process

The production of microspheres can be costly due to the intricate and specialized processes involved in their fabrication. Achieving precise control over factors such as size, drug loading, and encapsulation techniques requires advanced equipment and highly controlled conditions, which contribute to higher manufacturing costs. Additionally, scaling up from laboratory-scale production to industrial-scale manufacturing presents significant challenges. These challenges include maintaining the uniformity and quality of the microspheres, ensuring consistent drug encapsulation, and overcoming technical hurdles related to batch production. The complexities of scaling up, coupled with the associated costs, make the industrial production of microspheres a costly and technically demanding endeavor [16, 17].

Drug Loading and Release Challenges

Achieving consistent drug loading and controlling drug release are key challenges in the development of drug-loaded microspheres. Inconsistent drug loading can arise due to factors like variations in the microsphere fabrication process, polymer composition, and solvent evaporation rates, leading to uneven drug distribution [18]. This results in fluctuating release rates and can compromise therapeutic efficacy by causing inadequate drug concentrations or potential side effects. Similarly, maintaining a specific and controlled release profile is difficult, as factors such as the drug's properties, microsphere material, and environmental conditions all influence release behavior. In some cases, the drug may release too quickly or too slowly, leading to either burst release or suboptimal exposure at the target site. The challenge is further compounded by potential drug-polymer interactions and microsphere degradation, making the design of stable, controlled release systems a complex and highly optimized process [19].

Biocompatibility and Safety Concerns

Biocompatibility and safety are crucial considerations in the development of drug-loaded microspheres, as the materials used must not only deliver the drug effectively but also ensure the safety of the patient. One significant concern is toxicity, as certain synthetic polymers used in the formulation of microspheres may provoke toxic reactions or inflammatory responses in the body. This can occur, especially when the materials are not fully biocompatible or if they degrade in ways that result in the release of harmful substances [20]. Inadequate biocompatibility can lead to immune system activation, tissue irritation, or even more severe systemic toxicity. Another critical issue is the degradation of the microsphere material over time. As the microspheres break down, they can release byproducts or residues that may be harmful to the body. These degradation products, depending on their nature, could cause adverse effects such as organ toxicity, prolonged inflammation, or impaired

healing. Furthermore, the accumulation of such byproducts in the body over time can pose risks to long-term safety, potentially leading to chronic health issues. Therefore, careful selection of biocompatible materials and thorough evaluation of degradation pathways are essential to mitigate these risks and ensure that the drug delivery system remains safe throughout its use [21].

Risk of Migration

The risk of migration is a major concern with drug-loaded microspheres, especially in embolic therapies where they are used to block blood flow to specific areas, such as tumors. If the microspheres migrate from the intended site, they can cause embolism, blocking blood vessels in unintended locations and leading to serious complications like organ ischemia or tissue damage. Additionally, migration may result in microspheres lodging in critical organs such as the lungs, liver, or kidneys. Poor control over microsphere distribution can further exacerbate the issue, causing uneven drug delivery, which reduces therapeutic effectiveness or causes harm. This underscores the importance of precise formulation and careful administration to ensure microspheres stay localized and provide safe, effective treatment [22].

Limited Stability

The shelf life and storage conditions of drug-loaded microspheres are critical challenges due to the limited stability of certain formulations, particularly those containing sensitive drugs. Environmental factors like temperature, light, and humidity can degrade the drug and affect the microsphere's performance, necessitating specific storage conditions such as refrigeration or controlled environments. These requirements complicate manufacturing, distribution, and increase costs, limiting accessibility. Moreover, improper handling or deviations from optimal storage conditions can compromise the drug's efficacy and safety. Therefore, improving the stability of microsphere formulations and finding more flexible, cost-effective storage solutions is essential for their long-term use and clinical effectiveness [23, 24].

Biological Barriers

Biological barriers present significant challenges for drug-loaded microspheres, particularly in tissue penetration and immune system recognition. Microspheres may struggle to reach target tissues, especially in areas with protective barriers like the blood-brain barrier or tumor microenvironments, limiting their effectiveness. Additionally, the immune system may identify microspheres as foreign bodies, leading to premature clearance and reduced drug release time, which diminishes therapeutic efficacy. Immune responses can also cause inflammation and other adverse effects, further complicating treatment. Overcoming these barriers requires careful formulation design to improve tissue penetration, minimize immune recognition, and enhance delivery effectiveness [25].

Regulatory and Approval Challenges

Regulatory hurdles present significant challenges in the development and approval of microsphere-based products. The process is complex and time-consuming, requiring extensive preclinical and clinical testing to demonstrate safety, efficacy, and quality. Regulatory agencies, such as the FDA or EMA, demand detailed documentation and evaluations of both the formulation and manufacturing processes. Additionally, compliance with regulatory requirements incurs substantial costs, including expenses for clinical trials, validation, and quality control. These financial burdens can hinder the commercialization of microsphere-based therapies, making regulatory approval a major obstacle in their development [26, 27].

Potential for Unintended Interactions

Unintended interactions with drug-loaded microspheres can affect the release and activity of other medications in the body. The microsphere material may influence drug metabolism or absorption, leading to reduced efficacy or increased toxicity of concurrent treatments. Additionally, variability in the microsphere formulation can cause bioavailability issues, with drugs being released too quickly or slowly, impacting therapeutic outcomes. To ensure safety and efficacy, careful formulation of microspheres and thorough assessment of potential drug interactions are essential [28].

Difficulty in Patient Customization

Patient customization is a major challenge in the development and use of drug-loaded microspheres, particularly when trying to tailor formulations to meet the specific needs of individual patients. Personalizing microsphere-based therapies can be especially complex for drugs that require highly precise release profiles or targeted delivery. Variations in individual patient physiology, such as differences in metabolism, organ function, or disease states, can affect how the microspheres interact with the body and influence drug release and distribution. Adjusting the release rate to align with a patient's specific therapeutic needs is especially difficult for drugs with narrow therapeutic windows or those requiring sustained, controlled delivery. Customizing microsphere formulations often involves intricate modifications, such as changing the polymer type, adjusting drug loading, or altering the matrix composition. However,

these adjustments are not always practical or cost-effective. Furthermore, creating a universal method for customizing microspheres remains a challenge, as personalized treatments require extensive testing, additional resources, and specialized manufacturing processes. As a result, offering truly personalized microsphere-based therapies for each patient is complex, underscoring the need for continued innovation in formulation techniques and drug delivery systems [6, 28].

Types and Mechanism of Drug Release

The release of drugs from microparticulate systems, which may include one or more distinct excipients, can be facilitated through a combination of specialized manufacturing methods, dissolution/diffusion processes, osmotic pressure-driven release, erosion, and other mechanisms. Typically, these mechanisms operate simultaneously, with one often playing a more dominant role in the drug release process. When the active pharmaceutical ingredient (API) is incorporated into a polymer matrix, the dissolution behavior of the polymer system becomes a critical factor in the microparticulate release. This behavior, however, is influenced by multiple factors, including the properties of the drug, the formulation, and the characteristics of the release medium [29].

Micrometrics

When a polymer matrix is used, the active ingredient can diffuse through either the water-filled pores or the continuous polymer network as shown in Fig. 1. Drugs soluble in water may dissolve within these aqueous pore networks. The polymer chains swell upon water absorption, leading to the formation of additional pores and/or increased osmotic pressure. This swelling process allows more pharmaceutical molecules to enter the aqueous phase, which in turn raises the drug's effective diffusion coefficient and increases the overall volume. Additionally, the polymer matrix may undergo erosion, either at its bulk or surface, further influencing drug release [30].

Microcapsules

The behavior of the film-forming polymer in microencapsulated or polymer-coated microparticles can change based on how it interacts with the surrounding media as shown in Fig. 2. Drug release mostly happens via the active ingredient diffusing through the polymer if it dissolves in the medium. However, osmotic effects must also be taken into consideration in the drug release process if the polymer forms a water-insoluble, permeable, or semipermeable membrane. Additionally, by introducing pores into the coating and hastening the release of the active ingredient, the use of water-soluble pore formers can improve the dissolution profile [31, 32].

Types of Microspheres

Bioadhesive Microspheres

Bioadhesive microspheres are advanced drug delivery systems that include microparticles and microcapsules with a drug core, ranging in diameter from 1 to 1000 µm. These microspheres are either entirely composed of bioadhesive polymers or feature an outer coating made from such polymers. The integration of bioadhesive properties into microspheres offers several advantages, including improved drug absorption and enhanced bioavailability. This is achieved through a high surface-to-volume ratio, which facilitates intimate contact with the mucus layer. Additionally, bioadhesive microspheres can be engineered for targeted drug delivery by incorporating plant lectins, bacterial adhesins, antibodies, and other targeting agents on their surface [33,

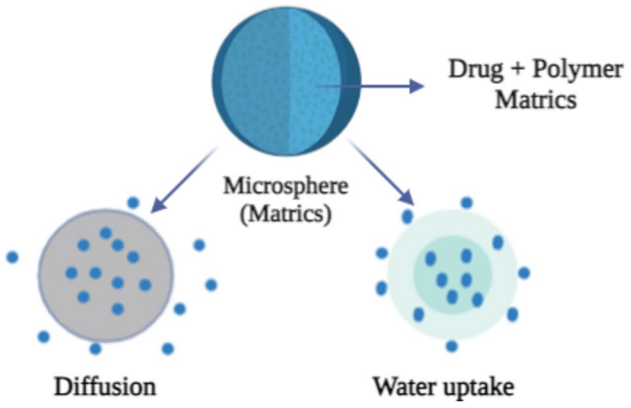


Fig. 1 Micrometrics

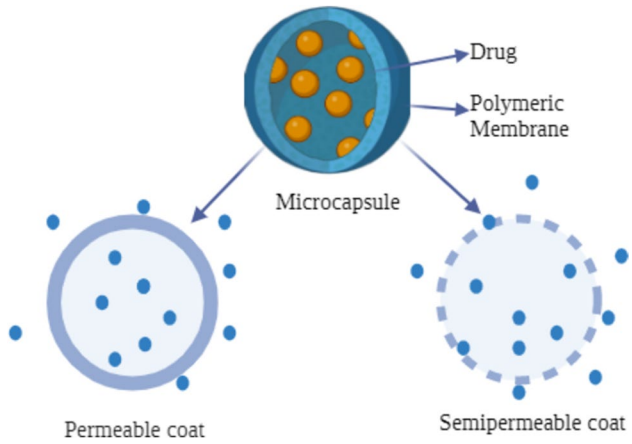


Fig. 2 Structure of microcapsule

34]. Overall, microspheres have significant potential for controlled and targeted drug release applications.

It is possible to tailor bioadhesive microspheres so that they stick to different mucosal tissues, such as those in the gastrointestinal tract, nasal cavity, urinary tract, and eyes. This adaptability allows for controlled drug release in both localized and systemic areas. Bioadhesive microspheres are injected into the mucosal tissues of the colon, stomach lining, and eye cavity to deliver drugs locally. By lowering the frequency of administration and offering prolonged drug release, these applications can greatly improve patient compliance. For instance, intranasal applications benefit from reduced mucociliary clearance, allowing for more effective and prolonged drug retention in the nasal mucosa, and sustained release formulations for ocular use can lessen the need for frequent eye drops [35]. The M cells in Peyer's patches of the gastrointestinal (GI) mucosa are known for their ability to selectively absorb microspheres composed of bioadhesive and bioerodible polymers. This property has been harnessed for various applications, including the delivery of antigens for immunization, plasmid DNA for gene therapy, and protein and peptide medications. Additionally, bioadhesive microspheres enhance the oral bioavailability and absorption of drugs such as furosemide and riboflavin by maintaining their presence in the absorption window of the GI mucosa. Another notable application of bioadhesive microspheres is in mucosal immunization, where they offer a non-invasive, single-shot vaccination approach by providing controlled and sustained release of antigens [36]. Designed to adhere to mucosal tissues, these microspheres enhance drug residence time and absorption, making them ideal for gastrointestinal, ocular, and intranasal drug delivery. The mechanism of action involves intimate adhesion to mucosal surfaces, enhancing drug residence time and absorption. Recent advancements include bioadhesive microspheres loaded with antigens for mucosal vaccination, demonstrating improved immune responses. These microspheres have been applied in therapies for gastrointestinal infections, where enhanced bioavailability and patient compliance are key advantages [37].

Different Theories Explaining the Mechanism of Bio-adhesion

Electronic Theory According to the theory, due to differences in their electronic structures, an adhesive polymer and a mucus glycoprotein network will transfer electrons when they come into contact. According to this interaction, the interface develops an electrical double layer that charges when in contact and discharges when the materials are separated. The electrical charges of the mucoadhesive material and the biological substrate are opposite, which causes electron transfer and the creation of an electrical double layer, which in turn controls the mucoadhesion strength [38, 39].

Adsorption Theory According to this theory, surface forces between atoms cause materials to stick together after first coming into contact. This theory states that after their initial interaction, the forces between the two surfaces will cause the mucoadhesive material to adhere to the biological surface. Weak forces, like van der Waals interactions, are important at the interface in the context of adsorption. Primary and secondary types are distinctions made between chemical bonds. Strong and covalent primary chemical bonds result in the creation of long-lasting bonds, which is why they are typically unwanted in bio-adhesion. On the other hand, weaker secondary chemical bonds such as those resulting from hydrophobic, hydrogen, van der Waals, and electrostatic interactions help to create the transient adhesion required for muco-adhesion [39].

Wetting Theory The adherence of liquids or low-viscosity bioadhesives to biological surfaces is fully explained by this theory. It describes adhesion in bioadhesives used in drug delivery using ideas of surface energies and interfacial tensions. The energy released per square centimeter when an interface forms is referred to as the "work of adhesion". The contact angle and the thermodynamic work of adhesion are both included in the wetting theory, which provides information about the adhesive characteristics of bioadhesives [39].

Diffusion Theory This theory states that the polymer chains attach themselves to the mucus and deeply enough to form a semipermanent adhesive bond. Glycoprotein, which is present in the mucus membrane, and the bioadhesive material interact closely. The polymer chains pierce the mucus; the precise depth of penetration required to produce adequate bioadhesion is contingent upon the diffusion coefficient, duration of contact, and additional experimental factors. When cross-link density rises, the diffusion coefficient, which is dependent on molecular weight, rapidly diminishes. This implies that crucial factors influencing inter-diffusion control are the flexibility and chain segment mobility of the mucus glycoprotein molecules and the bioadhesive polymer. A concentration gradient is created during chain interpenetration [39, 40].

Magnetic Microspheres

As an alternative to conventional radiation techniques, which employ extremely penetrating radiation that is absorbed throughout the body, consider using magnetic microspheres. Its toxicity and side effects restrict its use. Applications for magnetic radioactive microspheres are comparable to those

for non-radioactive spheres. The target site is the direction of a magnet that is externally positioned. The magnet can be housed in apparatus that resembles an open magnetic resonance imaging scanner, or it can be a permanent magnet in the shape of a rod of any length. After being inserted into a blood vessel, the loaded microspheres can accumulate at the target location in as little as 30-min and release radiation that destroys nearby cancer cells. It usually takes a few days or weeks for the therapeutic effect to take effect, depending on the substance [41]. Microspheres must be embedded with microscopic magnetic particles, such as iron, in applications that require in vivo magnetic targeting, such as magnetic drug delivery, to guarantee that the particles are drawn to a magnetic field. To enable efficient external magnetic guidance within the vascular system, these magnetic carriers should have a suitable size range of 200 nm to 3 mm and exhibit high magnetization. Because they are easier to capture and have better targeting capabilities than nanospheres, microspheres that are 1–2 mm in diameter are generally preferable for such uses [42, 43]. These microspheres localize drugs using an external magnetic field, offering targeted therapy for cancers like glioblastoma and enhancing localized chemotherapy efficacy. Employed in imaging techniques like MRI for enhanced contrast and localization, especially in detecting cancerous tissues. The mechanism relies on external magnetic fields to localize the microspheres at specific sites, enabling targeted drug delivery. Recent studies have demonstrated the application of magnetic microspheres in glioblastoma treatment, where localized chemotherapy minimizes systemic toxicity [44]. Their advantages include reduced side effects and enhanced drug concentration at the target site.

Principle of Magnetic Microspheres for Drug Targeting

"Drug targeting" refers to a specialized method of drug delivery that aims to precisely target the site of action or absorption, which could be a particular cell, organ, or tissue. Targeted specificity's main goal is to increase drug delivery effectiveness while reducing toxicity and adverse effects. The idea that a drug can be encapsulated within a magnetic microsphere or conjugated to its surface forms the basis of the magnetic drug transport technique as shown in Fig. 3. When the magnetic carrier is injected intravenously, it gathers in the magnetic field-applied region, and magnetic agglomeration frequently improves its localization even more [45].

Targeted medication delivery is made possible by the buildup of magnetic carriers at the target location. The effectiveness of this accumulation is dependent on a number of physiological factors, including blood flow rate, magnetic field strength, surface features, and particle size. The carriers' extravasation into the intended area is aided by the magnetic field. This method minimizes the harmful effects on the surrounding healthy tissues and the body as a whole by enabling the achievement of very high concentrations of chemotherapeutic agents close to the target site. As a result, compared to traditional techniques, considerably higher drug concentrations can be delivered to specific disease sites [11, 46].

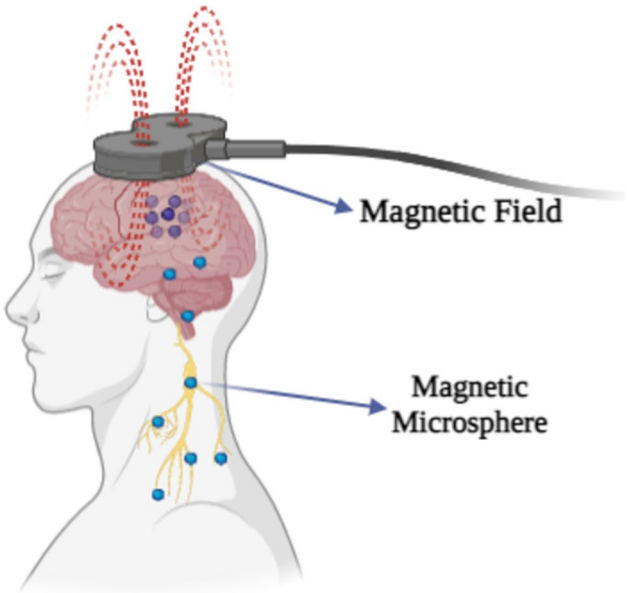


Fig. 3 Magnetic microsphere

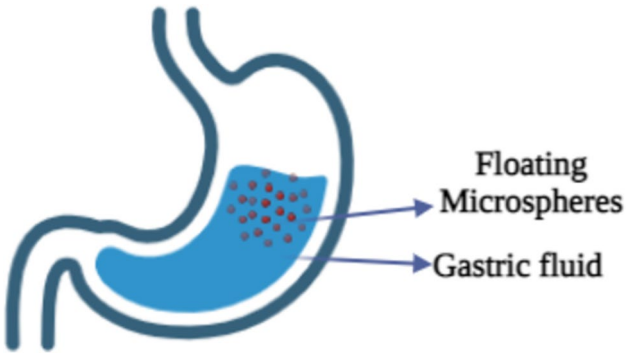


Fig. 4 Floating microsphere

Floating Microspheres

Drug delivery systems that are hydrodynamically balanced or floating can float in the stomach for longer periods of time without slowing down the pace of gastric emptying because they have a lower bulk density than gastric fluids. These floating systems slowly empty the stomach while releasing the medication at a regulated rate as shown in Fig. 4. As a result, this method increases the amount of time that drugs

remain in the stomach and improves the management of plasma drug concentrations.

Floating microspheres are spherical, empty particles devoid of a core. By employing an emulsion solvent evaporation process, these free-flowing particles whose sizes range from 1 to 1000μm have created non-effervescent hollow polycarbonate microspheres. This preparation, which is controlled by gastrointestinal transit, is intended to float on gastric juice that has a specific density lower than one. Transit through the stomach is delayed as a result of this characteristic. When the drug is released gradually at the intended rate, there are less fluctuations in the plasma drug concentration and more gastric retention [47, 48]. Utilized for sustained drug release in gastric conditions, they float on gastric fluids, providing prolonged retention and controlled drug delivery for antacids and antibiotics.

Mechanism of Floating Microspheres

The gel-forming polysaccharides and polymers inside the microspheres hydrate when exposed to stomach fluid, creating a colloidal gel barrier. The drug release rate is regulated by this gel layer, which also modifies the rate at which fluid enters the system. The hydration of the underlying hydrocolloid keeps the gel layer stable as the outer surface of the dosage form dissolves. Moreover, the air that is trapped inside the enlarged polymer lowers the overall density of the microspheres, increasing their buoyancy. Nonetheless, in order to preserve appropriate buoyancy and guarantee that the delivery system operates as intended, a minimum amount of gastric content is necessary [49].

Radioactive Microspheres

Radiation immobilization therapy uses microspheres, which are larger than capillaries and have a size range of 10–30 nm. The arteries supplying the intended tumor are injected with them. These radioactive microspheres do not harm healthy tissues; instead, they concentrate high radiation doses on particular areas. Radioactive microspheres can be classified into three types: those that emit α, β, and γ. Radioactive microspheres are tiny, spherical particles that are impregnated with or coated with radioactive isotopes. They generally range from a few micrometers to several millimeters in diameter [50, 51].

Microspheres can be fabricated from a variety of materials, each chosen for its unique properties that make it suitable for specific applications. Glass microspheres are often preferred for their ability to be precisely engineered, particularly in medical applications where they are used to contain and release radioactive material in a controlled manner. Polymers, both biodegradable and non-biodegradable, are another common choice, offering flexibility in drug delivery systems due to their ability to be designed for controlled release over time. Biodegradable polymers, in particular, break down within the body, releasing their contents gradually while avoiding long-term accumulation, while non-biodegradable polymers may provide more sustained, long-term delivery [52]. Ceramics are also used in microsphere production, prized for their stability and ability to provide specific release profiles. These materials can be engineered to release substances, such as drugs or radioactive elements, in a controlled manner, making them essential in fields such as targeted therapy or diagnostics. The choice of material depends on the desired release characteristics, stability, and the specific application for which the microspheres are intended [51]. Used in embolization for treating liver and other cancers, these microspheres deliver localized radiation, minimizing systemic toxicity. Utilized in tracing and imaging for organ-specific studies, helping in precise diagnostics for conditions like liver dysfunctions.

Mechanism of Radioactive Microspheres

Radiation emission is a key feature of radioactive microspheres, with the isotopes inside them decaying over time and emitting radiation in the form of alpha, beta, or gamma rays as shown in Fig. 5. In medical applications, this radiation is utilized to target and treat specific areas of the body, such as tumors, by delivering localized doses of radiation. Similarly, in industrial applications, the radiation emitted from the microspheres can be used to trace and monitor processes. The targeting and localization of microspheres are

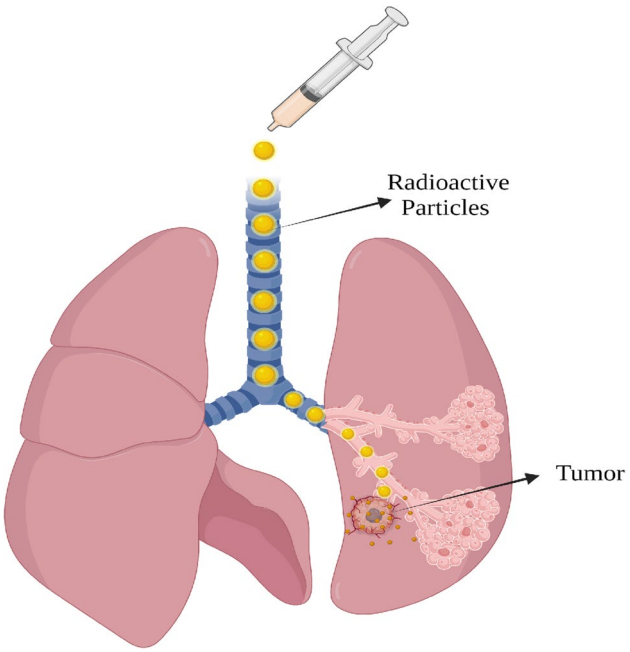


Fig. 5 Radioactive microsphere

particularly important in medical treatments, where they are often designed to selectively target specific tissues or organs. This is achieved through the microsphere's size and its ability to be selectively absorbed by certain tissues, ensuring that the radiation is delivered precisely where it is needed for therapeutic purposes. This targeted approach improves the effectiveness of treatment while minimizing damage to surrounding healthy tissue [52, 53].

Polymeric Microspheres

The different types of polymeric microspheres can be classified as:

Biodegradable Polymeric Microspheres

Drug delivery systems use natural polymers like starch because of their biodegradability, biocompatibility, and bioadhesive qualities. These biodegradable polymers swell significantly in aqueous environments, which facilitates their long-term adhesion and gel formation on mucous membranes. The polymer's concentration affects the drug release's rate and extent, enabling regulated and continuous drug delivery over time. The complex drug loading performance of biodegradable microspheres poses a notable challenge in clinical settings. It may be challenging to precisely control drug release due to this complexity. Notwithstanding this limitation, the advantageous properties of biodegradable polymers allow for a wide range of applications in microsphere-based therapies [18, 54]. These microspheres, particularly biodegradable types, are applied in cancer immunotherapy and chronic disease management due to their controlled release and biocompatibility. Biodegradable polymeric microspheres utilize hydrolytic or enzymatic degradation mechanisms for controlled drug release. Recent applications include cancer immunotherapy, where microspheres encapsulate checkpoint inhibitors for localized delivery. Their advantages include biocompatibility, reduced toxicity, and versatility in encapsulating diverse therapeutic agents [55].

Synthetic Polymeric Microspheres

Clinical applications make extensive use of synthetic polymeric microspheres, which are used as drug delivery vehicles, embolic particles, bulking agents, fillers, and more. Their biocompatibility and safety are widely acknowledged. These microspheres' tendency to migrate from the injection site, which raises the possibility of embolism and possible harm to other organs, is a serious disadvantage [56].

Classification of Polymer Used in Preparation of Microspheres

The tables presented provide a comprehensive classification of polymers utilized in the preparation of microspheres, highlighting their physicochemical properties, applications, and benefits. Table 1 categorizes proteins and waxes derived from plant or animal origins, such as albumin, gelatin, and casein, emphasizing their role in drug delivery through properties like solubility, crosslinking potential, and stability. Table 2 outlines polysaccharides, including sodium hyaluronate, starch, and chitosan, which are pivotal in controlled release, mucoadhesive systems, and colon-targeted delivery. Table 3 discusses cellulose derivatives like methyl cellulose and hydroxypropyl methyl cellulose, known for their gelation, film-forming, and pH-independent solubility. Table 4 delves into synthetic polymers such as PLA, PLGA, and polymethacrylates, valued for their biodegradability, pH-responsive behavior, and targeted delivery applications. Together, these tables underscore the versatility of polymers in optimizing microsphere-based drug delivery systems across diverse therapeutic and industrial applications [57].

Table 1 Proteins and waxes of plant or animal origin

Polymer	Physicochemical properties	Application and benefits
Albumin [58]	Exhibits high solubility in aqueous environments, carries a net negative charge at physiological pH (7.4).	Facilitates pulmonary delivery of therapeutics. Utilized in combination with alginate for controlled drug release.
Gelatin [59]	Displays amphoteric nature. Exists in two forms: Gelatin A with an isoelectric point (IEP) ranging from pH 7 to 9.4, and Gelatin B with an IEP ranging from pH 4.8 to 5.5. It initially swells before dissolving in water.	In acidic conditions, gelatin coacervates with negatively charged polymers and exhibits high crosslinking potential, functioning as an emulsifier, stabilizer, and binder. It is capable of thermoreversible gelling and its properties vary with pH, affecting swelling, dissolution, and erosion.
Casein [60, 61]	Characterized by its hydrophilic nature and metal-binding capacity. Shows insolubility in water at its isoelectric point (pH 4.6).	Used in the form of calcium caseinate, it demonstrates reversible thermal gelation, enhancing its solubility and functionality for various biomedical applications.

Table 2 Polysaccharides of various origin

Polymer	Physicochemical properties	Application and benefits
Sodium hyaluronate [62, 63]	Soluble in water, exhibits high viscosity even at low concentrations.	Utilized in microspheres for nasal, vaginal, and ophthalmic delivery systems.
Starch [64, 65]	Soluble in hot water following a period of gelatinization.	Applied in spray drying, extrusion, molecular inclusion, and coacervation with proteins. Forms hydrocolloids for controlled release via swelling, diffusion, and erosion.
Chitosan [66, 67]	Soluble in weak acids, possesses mucoadhesive properties, and interacts with negatively charged surfaces.	Engages in ionotropic gelation, coacervation with anions, and modified emulsification processes.
Guar gum [68, 69]	Water-soluble, nonionic galactomannan forming a thixotropic solution; stable in pH 4–10.5.	Employed for controlled release, colon-targeted drug delivery, and as an appetite suppressant with thermo-reversible properties.
Sodium alginate [70, 71]	An anionic polymer that dissolves in water, it can coacervate with polycations such as poly-L-lysine or chitosan, as well as divalent ions like Ca^{2}+, Sr^{2+}, and Ba^{2+}.	Functions as a disintegrants and binder with high swelling capacity. Exhibits pH-dependent swelling, is nontoxic and nonirritant, and possesses low density, enabling flotation in gastric fluids.
Tragacanth [72, 73]	It is soluble in water, has arabinogalactan and bassorin that can swell, and it swells quickly in water.	Acid-resistant emulsifier, used for oil encapsulation and as a stabilizer in emulsions.
Pectin [74, 75]	Negatively charged polymer. Gelation is influenced by the degree of esterification, cation concentration ($Ca2^{+}$, $Zn2^{+}$), temperature, and pH.	Utilized for in situ gelling, sustained delivery, and as a dietary fiber.
Xanthan gum [76, 77]	Soluble in both warm and cold water; unaffected by pH. Anionic polyelectrolyte.	Does not gel at room temperature but allows cryogelation. Provides stable viscosity across a wide pH range, serving as an emulsion stabilizer and for controlled release, including colon-targeted delivery.

Method of Preparation

Single Emulsion Techniques

Prepare different types of proteins and carbohydrates using this method. First, natural polymers are dissolved in an aqueous phase after this aqueous phase is distributed into a non-aqueous phase, often referred to as the oil phase Heat-sensitive materials should not be subjected to heat denaturation as shown in Figure 6. If chemical cross-linking is added during manufacturing and is subsequently centrifuged, cleaned, and separated, it has the drawback of exposing the active pharmaceutical ingredient to excessive amounts of chemicals [103, 104].

The single emulsion technique is a widely used method for preparing microspheres, particularly for encapsulating hydrophobic drugs. Its advantages include simplicity, cost-effectiveness, and suitability for heat-sensitive compounds, making it an accessible option for many pharmaceutical applications. This method allows for better control over particle size and drug distribution within the microsphere matrix. However, it also has certain disadvantages, such as low encapsulation efficiency for hydrophilic drugs, and it is highly sensitive to process parameters like stirring speed, solvent type, and polymer concentration. These factors can lead to inconsistent microsphere size and drug loading efficiency, which may affect the overall performance and reproducibility of the final product [104].

Double Emulsion Techniques

Using this technique, several emulsions are made, particularly a water-in-oil-in-water (W/O/W) emulsion. First, it is mixed into an aqueous polyvinyl alcohol solution to create a primary water in oil (W/O) emulsion. After that, the W/O/W emulsion is continuously stirred for half an hour. Water is added to the emulsion gradually over a 30-min period. Following preparation, microcapsules are gathered via filtration and vacuum-dried. This method works especially well for encapsulating vaccines, peptides, proteins, and medications that dissolve in water as shown in Fig. 7 [105]. In this procedure, polymers that are synthetic or natural can be used. The aqueous protein solution with the active ingredients is dispersed using a continuous organic lipophilic phase. This is a process that creates the primary emulsion by homogenizing or vigorously mixing an initial dispersion in the oil/organic phase. To make a multiple emulsion, polyvinyl alcohol (PVA) in an aqueous solution is then added. Denaturation, or hardening, is then brought about by the addition of a larger aqueous phase. The microspheres are then separated, cleaned, dried, and

Table 3 Cellulose derivatives

Polymer	Physicochemical properties	Application and benefits
Methyl cellulose [78, 79]	Soluble in cold water, amphiphilic with a gelation temperature (Tsol-gel) around 80°C.	Functions as an emulsifier; forms pseudoplastic solutions; its properties are independent ofpH.
Carboxymethyl cellulose sodium [80, 81]	Anionic cellulose ether, dispersible in water (forms a colloidal solution)	Utilized in injectable, thermoreversible gel formulations; exhibits mucoadhesive properties.
Hydroxy propyl methyl cellulose [82, 83]	Water soluble, nonionic	Demonstrates reversible thermal gelation; used as an emulsion stabilizer with film-forming capabilities.
Ethyl cellulose [84, 85]	Water-insoluble, provides a hydrophobic coating.	Employed for its film-forming ability; enables membrane-controlled diffusion and modified drug release in gastroretentive systems.
Hydroxypropyl cellulose [82, 86]	Soluble in cold water, compatible with waxes and oils.	Exhibits high surface activity; possesses excellent film-forming properties.
Cellulose acetate butyrate[87, 88]	Soluble in mixtures of acetone and water, insoluble in water.	Used for semipermeable coatings; aids in extended-release formulations, influencing diffusion and matrix erosion.

gathered using the multiple emulsion technique of oil in water in oil (o/w/o) [106, 107].

The double emulsion technique is widely used for encapsulating hydrophilic drugs, proteins, and peptides within microspheres. Its advantages include the ability to encapsulate both hydrophilic and hydrophobic drugs effectively, making it highly versatile for various pharmaceutical applications. This method allows for controlled and sustained drug release profiles and is particularly useful for delivering sensitive biological molecules. The technique also has disadvantages, such as being a complex and multi-step process that requires precise control over parameters like emulsification speed, solvent evaporation, and polymer concentration. It often results in low drug encapsulation efficiency and can lead to drug loss or denaturation during the preparation process. These limitations can affect the consistency, scalability, and cost-effectiveness of the final microsphere product [106].

Polymerization

Normal Polymerization

The first step in the bulk polymerization process usually involves heating a monomer or a mixture of monomers, catalyst, and initiator. One way to accomplish this is to incorporate the drug during polymerization, which will allow the resulting polymer to be formed into microspheres. While this method concentrates on the formation of pure polymers, it is difficult to control the heat produced by the reaction, which can negatively impact active ingredients that are heat-sensitive. On the other hand, lower temperatures are used in suspension polymerization, also known as pearl polymerization. Using this technique, the active medication is heated along with a mixture of monomers to create droplets that are distributed throughout a continuous aqueous phase [108].

The normal polymerization technique is commonly used for the preparation of microspheres by heating a monomer or a mixture of monomers with an initiator and catalyst, allowing polymerization to occur. Its advantages include the ability to produce high-purity microspheres with controlled particle size and morphology, making it suitable for various drug delivery applications. Additionally, this method allows for the incorporation of therapeutic agents during the polymerization process, ensuring uniform drug distribution within the polymer matrix. Normal polymerization also has notable disadvantages, such as the difficulty in controlling the exothermic reaction, which can lead to heat-induced degradation of sensitive drugs. Furthermore, the use of toxic monomers and organic solvents raises safety and environmental concerns, and the process can be time-consuming and challenging to scale up while maintaining product consistency [109].

Interfacial Polymerization

Using this method, the dispersed phase which is created when different monomers react at the interface between two immiscible liquid phases is enclosed by a polymer film. This process makes use of two reacting monomers: one is emulsified in an aqueous continuous phase, and the other is dissolved in the continuous phase [109, 110].

The interfacial polymerization technique is a widely used method for preparing microspheres, particularly for encapsulating hydrophilic and hydrophobic drugs within a polymer shell. Its advantages include the ability to form microspheres with precise particle size control and uniform coating, making it suitable for delivering a variety of therapeutic agents. The process occurs at the interface of two immiscible liquid

Table 4 Synthetic polymers

Polymer	Physicochemical properties	Application and benefits
Poly (lactic acid) (PLA) [89, 90]	It is water-insoluble and breaks down into CO_2 and H_2O over a 12- to 24-month period.	Exhibits biodegradability, making it suitable for prolonged release formulations in intramuscular (IM) or subcutaneous (SC) injections, implants, and oral solid dispersions.
Polylactic acid-glycolic acid copolymer (PLGA) [91, 92]	It is water-insoluble and breaks down into CO_2 and H_2O over a 12- to 24-month period.	Utilized in injectable or implantable systems (microparticles, gels) for both human and veterinary applications. It can exhibit pH-responsive or non-pH-responsive degradation and is used in bone tissue engineering.
Polyethylene glycols [93, 94]	Exists in liquid or solid states, depending on molecular weight (Mw).	Acts as a plasticizer in the walls of compressible microcapsules. Employed in oral insulin delivery and as a cell delivery medium when combined with other agents.
Polymethacrylates [95, 96]	Soluble in organic solvents; most are miscible with water.	Displays pH-dependent solubility and permeability, making it ideal for targeted gastric or enteric drug delivery systems.
Fumaryl diketopiperazine (FDKP) [97, 98]	At pH 6, it is highly soluble and has no biological activity.	Forms self-assembled, highly porous microparticles; used as a carrier in Technosphere® delivery systems.

phases, allowing for rapid polymerization and the creation of tailored microsphere structures with specific release profiles. The technique also has disadvantages, including the use of toxic monomers and organic solvents, which can pose safety and environmental concerns. Additionally, the polymerization reaction requires strict control over reaction parameters, such as temperature, agitation speed, and reactant concentrations, to ensure reproducibility and avoid structural defects in the microspheres [109, 111]. These limitations can increase the complexity and cost of large-scale production.

Phase Separation Coacervation Technique

This process involves lowering the polymer's solubility in the organic phase in order to encourage the development of coacervates, a phase rich in polymers. First, drug particles are dispersed in a polymer solution in this process. The drug particles are encapsulated and the first polymer phase is separated by the introduction of an incompatible polymer. Polymer solidification is caused by the addition of a non-solvent as shown in Fig. 8. Butadiene has been used as the incompatible polymer in this technique to prepare microspheres of polylactic acid (PLA). Process variables are very important in this method because the distribution of the polymer film, particle size, and the possibility of agglomeration of the formed particles are all greatly affected by the rate at which coacervates form [112]. Stirring the suspension at the right speed is crucial to avoid agglomeration because during microsphere formation, the polymerized globules start to stick to one another. Good stirring prevents agglomerates from forming. These variables are important because they affect the kinetics of particle formation and the process does not reach a defined state of equilibrium [113].

The phase separation coacervation technique is a widely used method for preparing microspheres, especially for encapsulating sensitive bioactive compounds, proteins, and peptides. Its advantages include high drug loading capacity, excellent encapsulation efficiency, and the ability to maintain drug stability during the encapsulation process, making it suitable for both hydrophilic and hydrophobic drugs. It allows for tailored release profiles by adjusting polymer concentration and process parameters. However, the technique also has disadvantages, such as being highly sensitive to environmental factors like temperature, pH, and stirring speed, which can result in inconsistent microsphere size and drug distribution. The process often involves the use of organic solvents and cross-linking agents, raising concerns about residual toxicity and environmental safety. These factors make it a technically demanding and less scalable process for industrial applications [112, 114].

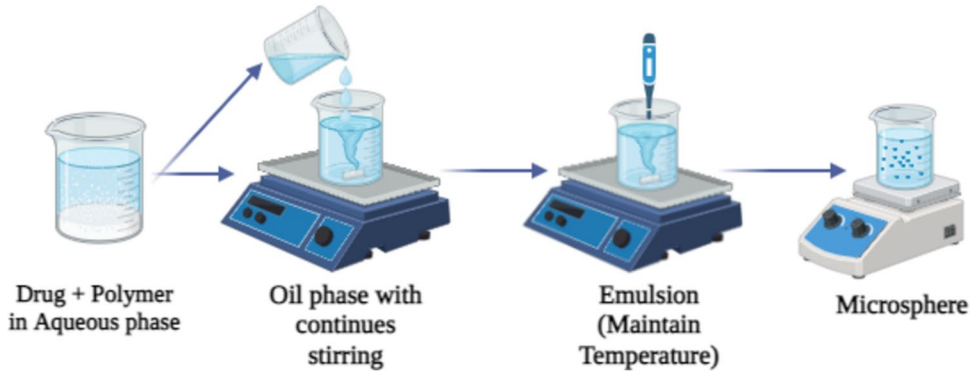


Fig. 6 Single emulsion techniques

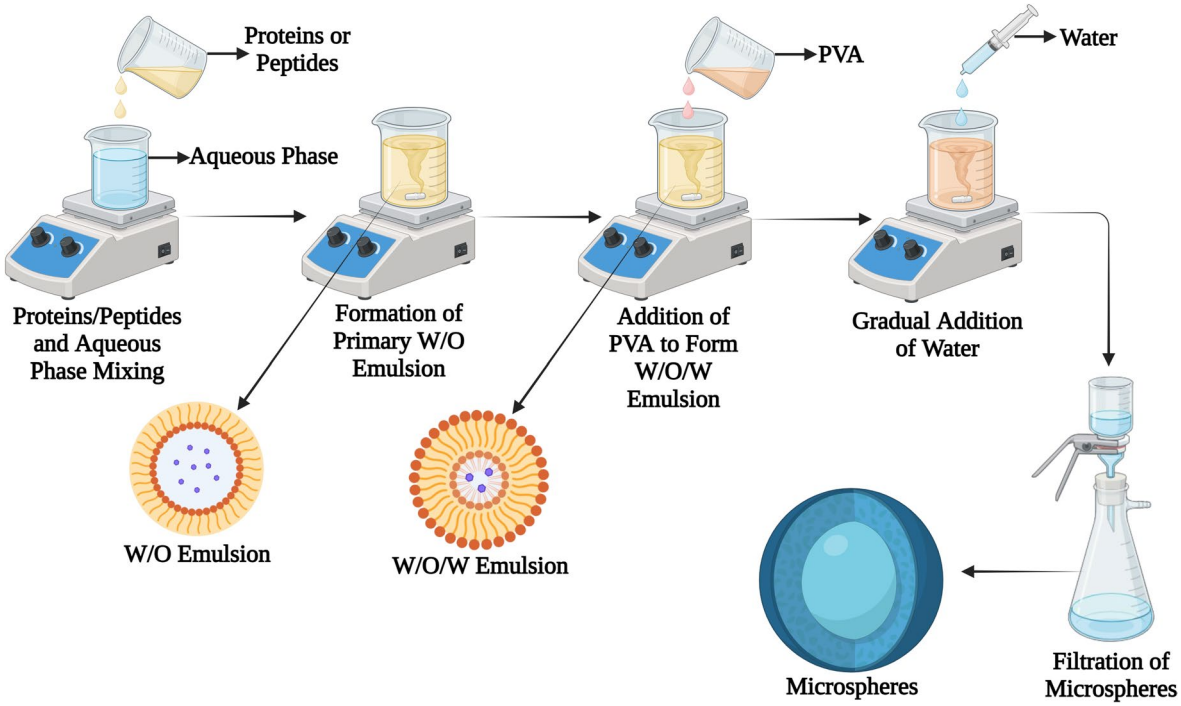


Fig. 7 Double emulsion techniques

Spray Drying

The polymer is first dissolved in an appropriate volatile organic solvent, like acetone or dichloromethane, before spray drying. A high-speed homogenization process is applied to the resultant solution to guarantee the compound's uniform dispersion. A hot air stream is then used to atomize this dispersion, producing tiny droplets or a mist as shown in Fig. 9. Microspheres with diameters ranging from one to one hundred micrometers are formed when the solvent evaporates quickly. The microparticles are separated from the hot air using a cyclone separator, and any remaining solvent is then purged using vacuum drying. This process's ability to create porous microparticles while operating in aseptic conditions is one of its main advantages [115, 116].

The spray-drying technique is a widely used method for preparing microspheres, particularly for heat-sensitive and moisture-sensitive drugs. Its advantages include rapid and scalable production, cost-effectiveness, and the ability to precisely control particle size and morphology. The process can be conducted under aseptic conditions, making it suitable for pharmaceutical applications, and it allows for encapsulation of both hydrophilic and hydrophobic drugs. The technique also has disadvantages, such as low encapsulation efficiency for water soluble drugs, potential drug degradation due to heat exposure, and the formation of porous or fragile microspheres if not properly optimized. The use of organic solvents may pose environmental and safety concerns, and maintaining uniform product quality on a large-scale can be challenging [117].

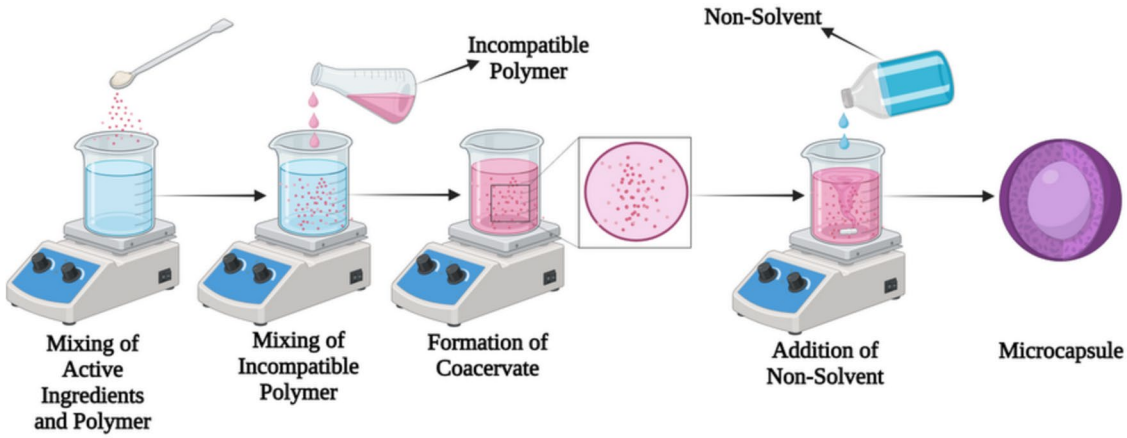


Fig. 8 Phase separation coacervation technique

Emulsion Crosslinking Method

The Emulsion Crosslinking Method is a sophisticated technique used to produce cross-linked polymer microspheres through the interaction of reactive functional groups within the polymers and aldehyde groups from cross-linking agents. The process begins with the preparation of a water-in-oil (W/O) emulsion, where an aqueous solution of the polymer is dispersed into an oily phase. This dispersion is stabilized by incorporating a suitable surfactant, such as Span 80 or dioctyl sodium sulfosuccinate, which helps to maintain the stability and uniformity of the aqueous droplets within the oil phase as shown in Fig. 10. Once a stable emulsion is achieved, the next step involves the introduction of a cross-linking agent, typically glutaraldehyde, into the emulsion. This cross-linker reacts with the polymer functional groups, facilitating the formation of a solid network within the droplets, thereby solidifying them into microspheres. After the cross-linking reaction is completed, the microspheres are subjected to filtration to separate them from the continuous oil phase. To remove any residual oil that might affect the properties of the microspheres, they are washed multiple times with organic solvents such as hexane or petroleum ether [118, 119]. Following this, the microspheres undergo a water wash to eliminate any remaining cross-linking agents, ensuring that the final product is free of unwanted chemical residues. The final step in the process involves drying the microspheres at room temperature for a period of 24 h. This comprehensive method allows for the creation of polymer microspheres with tailored properties, useful in a variety of applications including drug delivery systems, catalysis, and environmental remediation. The choice of surfactant, cross-linker, and washing solvents can be optimized depending on the specific requirements of the application, ensuring the desired characteristics and performance of the microspheres [120].

The emulsion crosslinking method for preparing microsponges offers several advantages, including the ability to control particle size and achieve high drug encapsulation efficiency, making it suitable for targeted drug delivery and sustained-release formulations. It also enables the creation of stable, porous structures that enhance drug loading and controlled release. This method has some disadvantages, such as the potential use of toxic crosslinking agents like glutaraldehyde, which may leave harmful residues, raising safety concerns. The process can be time-consuming and requires precise control over parameters such as stirring speed and crosslinking conditions to avoid particle aggregation and ensure uniformity [121]. These challenges can impact scalability and increase production costs.

Solvent Evaporation

The procedures are carried out in a liquid medium for production. A volatile solvent that doesn't mix with the production process' liquid phase is used to disperse the microcapsule during the manufacturing process. In a polymer coating solution, the core material meant for microencapsulation is either dissolved or dispersed. The desired microcapsule size is achieved by dispersing the core material mixture within the liquid production medium through agitation as shown in Fig. 11. The solvent is then removed from the polymer solution by heating the mixture, if possible. The polymer shrinks around the core material as the solvent evaporates, creating a microcapsule matrix. Either water-soluble or water-insoluble core materials may be employed. It is possible to obtain non-aqueous or aqueous (oil-in-water) formulations through the solvent evaporation process [122, 123].

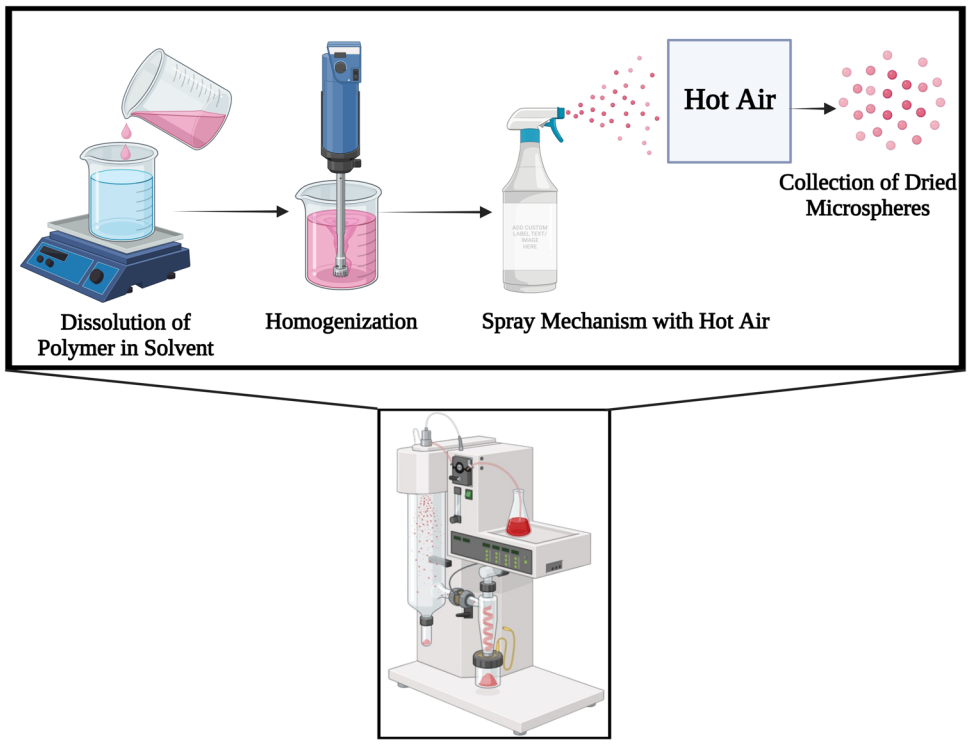


Fig. 9 Spray drying

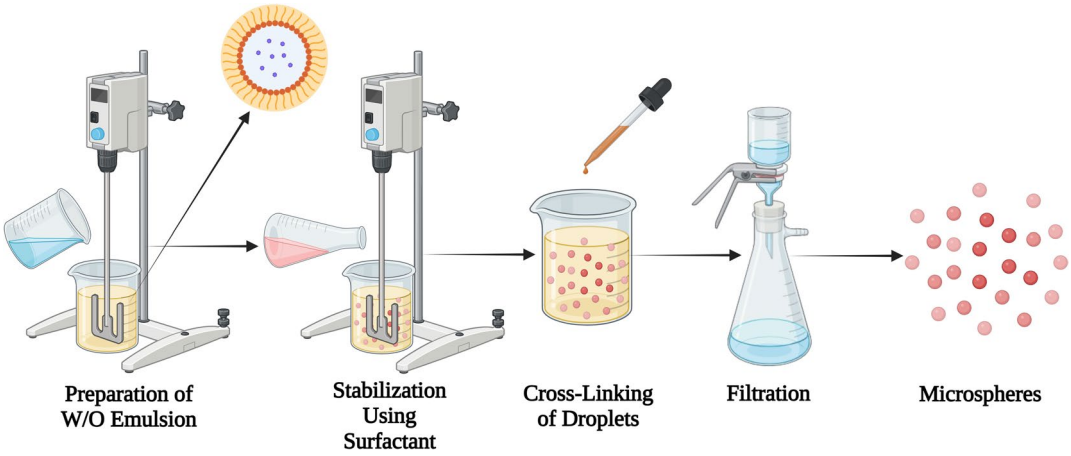


Fig. 10 Emulsion crosslinking method

The solvent evaporation method for preparing micro sponges is advantageous due to its simplicity, cost-effectiveness, and ability to produce uniform microspheres with controlled particle size and high encapsulation efficiency. It is particularly suitable for heat-sensitive drugs, as the process operates at moderate temperatures. It allows for the encapsulation of both hydrophilic and hydrophobic drugs, making it versatile for various applications. However, the method has disadvantages, including the potential for residual solvent toxicity, which may affect the safety of the final product. It requires careful control of process parameters, such as solvent removal rate and stirring speed, to prevent particle aggregation and ensure uniformity [121]. These limitations can complicate scalability and impact the environmental sustainability of the process (Tables 5, 6, and 7).

Ionic Gelation Method

This technique was employed to fabricate an alginate/chitosan particulate system designed specifically for controlled drug release applications. The process begins with the incorporation of the drug into an aqueous sodium alginate solution, creating a homogeneous mixture through continuous stirring. To facilitate the formation of microspheres, a

solution containing calcium ions ($Ca2^{+}$) and/or aluminum ions ($Al3^{+}$) is added dropwise to the alginate-drug solution. These ions act as cross-linking agents, causing the alginate to gel and form solid particulates around the drug. The resulting microspheres are then left in the original solution for a period of 24 h, allowing for internal jellification to ensure that the microspheres achieve the desired structural integrity and drug encapsulation. After this incubation period, the microspheres are separated from the solution through filtration as shown in Fig.12 [124, 125]. To ensure that the drug release profile meets the desired specifications, the release characteristics of the microspheres are evaluated across different pH environments. It is found that the complete release of the drug occurs at a pH range of 6.4–7.2, which is typically within the physiological pH range, making it suitable for in vivo applications. In contrast, the microspheres are designed to be resistant to release at acidic pH levels, which is advantageous for targeting specific sites in the gastrointestinal tract or other environments where an acidic condition is prevalent. This pH-responsive behavior helps in achieving controlled and targeted drug delivery, ensuring that the medication is released at the appropriate location and under suitable conditions for optimal therapeutic efficacy [126]. The use of alginate and chitosan in the particulate system not only provides a biocompatible matrix for drug encapsulation but also leverages the unique properties of these biopolymers to enhance the performance of the drug delivery system [127].

The ionic gelation method for preparing microsponges is advantageous due to its simplicity, mild processing conditions, and ability to encapsulate sensitive bioactive compounds like proteins and peptides without denaturation. It avoids the use of organic solvents and high temperatures, making it eco-friendly and suitable for biocompatible applications. This method allows for precise control of particle size and drug release profiles by adjusting the concentration of polymers and cross-linking agents. It has disadvantages such as the potential for variability in particle size and drug loading due to the sensitivity of the process to stirring speed and cross-linker concentration. The resulting microsponges may exhibit lower mechanical strength, and the need for specific ionic conditions can limit scalability and broader industrial application [128].

Evaluation Parameter of Microsphere

The characterization of the microparticulate carrier is an essential step in creating an efficient carrier for the delivery of proteins, medications, or antigens. The different microstructures that the microspheres display is important in determining the stability and release profile of the carrier.

Particle Size and Shape

Conventional Light Microscopy (LM) and Scanning Electron Microscopy (SEM) are commonly employed techniques to visualize and analyze microparticles.

Light Microscopy (LM)

Light microscopy is used to observe the overall shape and structure of microparticles. This technique provides insights into the outer structure and allows for the assessment of coating parameters in double-walled microspheres. By comparing images before and after coating, researchers can measure and analyze the changes in microsphere morphology. However, light microscopy has limitations in terms of resolution and may not capture fine surface details [133].

Scanning Electron Microscopy (SEM)

Scanning electron microscopy offers a higher resolution compared to light microscopy. SEM provides detailed

Fig. 11 Solvent evaporation

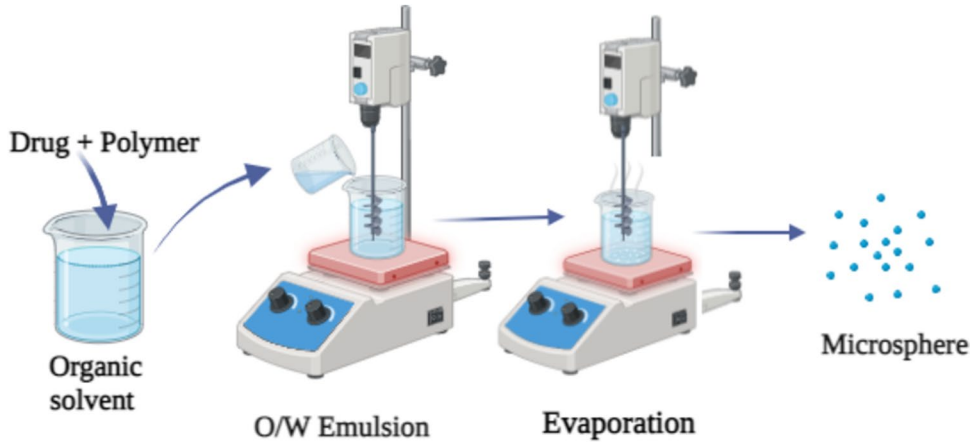


Table 5 Types of microspheres

Type	Description	Advantages	Applications	Examples
Bioadhesive microspheres [99]	Microspheres with adhesive properties to stick to biological surfaces, prolonging drug retention time.	Increased residence time at the target site; enhanced drug absorption.	Drug delivery to mucosal tissues like buccal, nasal, gastrointestinal linings.	Buccal delivery of antihypertensive drugs; nasal vaccines.
Magnetic Microspheres [100]	Microspheres containing magnetic materials, guided to the target site using an external magnetic field.	Precise drug targeting; reduces systemic side effects.	Cancer therapy, targeted drug delivery, and imaging.	Doxorubicin-loaded magnetic microspheres for tumor targeting.
Floating microspheres [101]	Low-density microspheres that remain buoyant in gastric fluids, increasing drug residence time.	Prolonged gastric retention; suitable for drugs absorbed in the stomach.	Treatment of gastric ulcers, gastroretentive drug delivery systems.	Floating microspheres of metformin for diabetes management.
Radioactive microspheres [102]	Microspheres loaded with radioactive isotopes for localized radiation therapy.	Minimizes damage to surrounding healthy tissues; highly effective for localized therapy.	Treatment of liver cancer and other localized tumors.	Yttrium-90 microspheres for selective internal radiation therapy.
Polymeric Microspheres [18]	Microspheres made from biodegradable or non-biodegradable polymers for sustained drug release.	Controlled and sustained drug delivery; high stability.	Delivery of anticancer agents, peptides, and proteins.	PLGA microspheres for cancer and vaccine delivery.

images of the surface structure of microspheres, revealing fine surface features and textures. When particles are cross-sectioned, SEM can also be utilized to investigate internal structures, such as double-walled configurations. This high-resolution imaging is crucial for a comprehensive understanding of the microspheres' morphology and structural integrity [134].

Surface Chemistry

Electron Spectroscopy for Chemical Analysis (ESCA), also known as X-ray Photoelectron Spectroscopy (XPS), is employed to analyze the surface chemistry of microspheres. ESCA provides detailed information about the elemental composition and chemical states of the surface atoms. By detecting photoelectrons emitted from the surface when irradiated with X-rays, ESCA helps in identifying functional groups and surface modifications. This analysis is essential for understanding how surface chemistry affects the interaction of microspheres with their environment and with biological systems [135].

Density Determination

Understanding the behavior of microspheres in different formulations and processing conditions requires an understanding of their density. The technique used to ascertain the density of microspheres is called multi-volume pycnometry. Using this technique, the volume of a known mass of microspheres in a liquid is measured, and the density is computed from these data. For microspheres to be handled and perform consistently in real-world applications, precise density determination is essential [136].

Isoelectric Point

An essential factor influencing microsphere stability and interactions with other materials is their isoelectric point (pI). Micro-electrophoresis is used to measure the electrophoretic mobility of the microspheres in order to determine the isoelectric point. This method entails putting a suspension of microspheres in an electric field and tracking the movement of the spheres. The pH at which the microspheres show no net charge is used to compute the isoelectric point. Comprehending the isoelectric point facilitates the enhancement of microsphere formulation and stability, especially in biological and pharmaceutical applications [137, 138].

Angle of Contact

An important metric for evaluating the wetting characteristics of microparticulate carriers is the contact angle. It is described as the angle that forms at the point of contact

Table 6 Preparation method of microspheres

Method	Description	Advantages	Disadvantages	Examples of use
Single emulsion [112, 121]	Involves dispersing a hydrophobic drug in an aqueous polymer solution, followed by solvent evaporation.	Simple, cost-effective, suitable for hydrophobic drugs.	Limited to hydrophobic drugs; sensitive to process parameters.	Controlled drug delivery for poorly water-soluble drugs.
Double emulsion [129, 130]	A water-in-oil-in-water (W/O/W) emulsion used to encapsulate hydrophilic drugs.	Effective for hydrophilic drugs and proteins; achieves sustained release.	Low encapsulation efficiency; complex and multi-step process.	Encapsulation of peptides, proteins, and vaccines.
Spray drying [131]	Atomizes a polymer-drug solution into hot air, leading to rapid evaporation and microsphere formation.	Scalable, fast production; suitable for heat-sensitive compounds.	Low efficiency for water-soluble drugs; potential drug degradation due to heat exposure.	Production of inhalable microspheres and sustained-release formulations.
Solvent evaporation [121]	Drug is dissolved in an organic solvent and emulsified in water; solvent is evaporated to form microspheres.	Versatile; allows for encapsulation of both hydrophilic and hydrophobic drugs.	Environmental concerns due to organic solvent usage; requires precise parameter control.	Delivery of anticancer agents, proteins, and antibiotics.
Interfacial polymerization [132]	Polymerization occurs at the interface of two immiscible liquids to form a polymer shell.	Precise particle size control; suitable for both hydrophilic and hydrophobic drugs.	Toxic solvents and monomers; complex process.	Encapsulation of vaccines, drugs, and imaging agents.
Phase separation coacervation [112]	Drug and polymer precipitate as a coacervate under specific conditions, forming microspheres.	High drug loading and encapsulation efficiency; suitable for sensitive bioactive compounds.	Requires precise parameter control; risk of agglomeration and irregular microsphere size.	Controlled release of proteins and hydrophobic drugs.
Emulsion crosslinking method [112]	Cross-linking occurs in an emulsion system where polymer droplets are stabilized and cross-linked chemically.	Suitable for natural polymers like chitosan; produces stable microspheres with controlled size.	Use of cross-linking agents may introduce toxicity; sensitive to process parameters.	Encapsulation of insulin, probiotics, and small hydrophilic drugs.
Ionic gelation [128]	Interaction between ionic polymers (e.g., alginate) and counterions forms microspheres.	Mild reaction conditions; preserves drug activity; biocompatible.	Poor mechanical strength and stability of microspheres.	Delivery of hydrophilic drugs and biologics like insulin.

Table 7 Summary of major applications of microspheres with advantages

Application	Example or uses	Key advantages
Therapeutics [190]	Drug delivery for cancer (e.g., glioblastoma, lung, and prostate cancer)	Targeted delivery, reduced systemic toxicity, controlled release
Vaccine delivery [191]	Microspheres encapsulating antigens for slow-release immunization	Prolonged antigen release, enhanced immune response
Diagnostics [192]	Eco-friendly dextran microbeads for multiplex assays	High sensitivity and specificity in biomarker detection
Tissue engineering [193]	Biodegradable microspheres as scaffolds for bone and cartilage regeneration	Promotes cell growth, provides structural support
Cosmetics [175]	Probiotics-loaded microspheres for skin formulations	Enhanced stability, improved viability of active ingredients
Environmental uses [194]	Microspheres for pollutant adsorption or controlled nutrient release in agriculture	Environmental safety, prolonged action, and biodegradability
Gene therapy [195]	Microspheres carrying DNA, RNA, or siRNA for genetic modifications	Protects genetic material, ensures targeted delivery
Ophthalmology [151]	Microspheres for sustained release of drugs to treat glaucoma and retinal diseases	Prolonged ocular drug retention, reduced administration frequency
Hormone replacement [196]	Microspheres for sustained delivery of hormones like estrogen and testosterone	Stable release, improved patient compliance
Antibiotic delivery [197]	Controlled delivery of antibiotics for infection management	Reduces drug resistance, localized delivery
Cardiology [198]	Microspheres for embolization therapy in heart diseases	Minimally invasive, localized treatment
Oncology [199]	Radioactive microspheres for brachytherapy and embolization in liver cancer	Localized radiation therapy, minimizes damage to healthy tissues

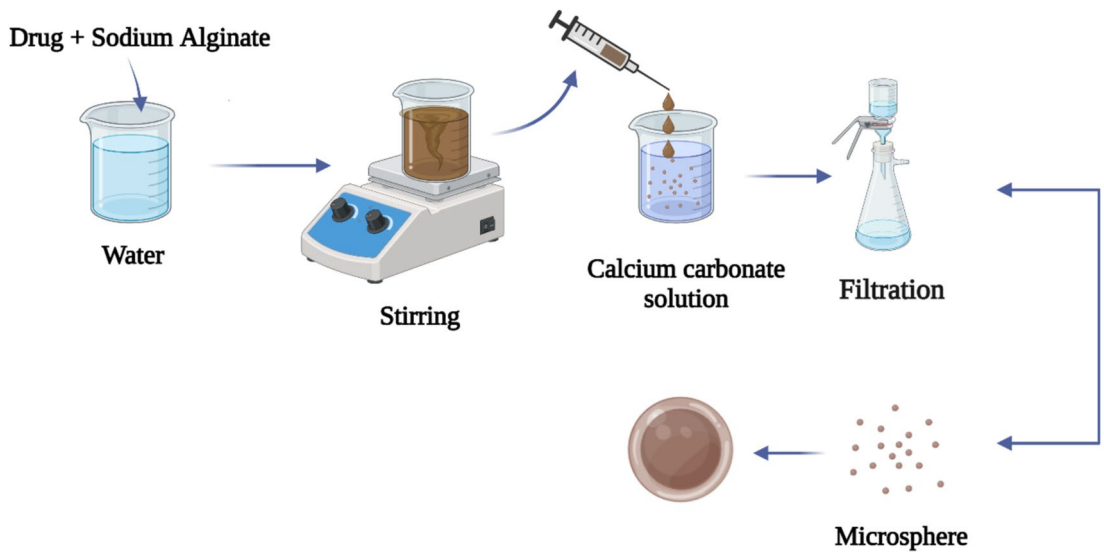


Fig. 12 Ionic gelation method

between the surface of the microsphere and the tangent to the liquid droplet. This viewpoint sheds light on how well the microsphere's surface interacts with different liquids. A goniometer or contact angle analyzer is used to measure the contact angle after a droplet of liquid is placed on the surface of a microsphere. The hydrophobic or hydrophilic character of the microsphere surface can be inferred from the contact angle. A hydrophilic surface is indicated by a low contact angle, whereas a high contact angle suggests a hydrophobic surface. For applications involving drug release, where the interplay between the microsphere and the surrounding medium can impact performance, an understanding of the contact angle is essential [139].

Drug Entrapment Efficiency

Measuring the effectiveness of a drug's incorporation into the microspheres during their preparation is known as drug entrapment efficiency. It is computed as the portion of the medication that stays inside the microspheres as a percentage of the total amount that was used at first. To extract the microspheres from the drug solution, lyse the microspheres in a buffer solution and then filter or centrifuge the lysate. This yields the entrapment efficiency. After that, the amount of the active medication is measured using the proper analytical techniques, like high-performance liquid chromatography (HPLC) [15]. The entrapment efficiency is calculated using the formula:

$$\text{EntrapmentEfficiency} = \frac{\text{Actual Drug Content}}{\textbf{TheoreticalDrugContent}} \times 100$$

Percentage Yield

Percentage yield measures the efficiency of the microsphere preparation process by comparing the weight of the final product to the total weight of the starting materials [15]. Percentage yield is calculated using the formula:

$$\text{Percentage Yield}(\%) = \left(\frac{\text{Weight of microspheres}}{\text{Total wieght of drug and polymer used}}\right) \times 100$$

Swelling Index

Swelling index provides information about the ability of the microspheres to absorb water and expand, which can impact drug release and stability. To determine the swelling index, microspheres are immersed in a suitable medium (usually water or a buffer solution) for a specified period. After immersion, the microspheres are removed, blotted to remove excess surface moisture, and weighed [140]. The swelling index is calculated using the formula:

$$\text{Swelling Index} = \left(\frac{\text{Mass of Swollen Microspheres} - \text{Mass of dry microspheres}}{\text{Mass of dry microspheres}}\right)$$

Flow Properties

Powder flow properties, such as bulk density, tapped density, Hausner's ratio, and angle of repose, are crucial parameters for evaluating the physical characteristics and handling behavior of microspheres. Bulk density measures the mass per unit volume of microspheres in their uncompressed state, influencing factors like flowability, packing, and drug release. Tapped density, determined after tapping the sample, provides insights into particle packing and porosity. Hausner's ratio, the ratio of tapped density to bulk density, indicates the flowability of microspheres, with lower values suggesting better flow properties. The angle of repose, the maximum angle formed by a pile of microspheres, is another indicator of flowability, with lower angles signifying better flow. By understanding and controlling these parameters, researchers and manufacturers can optimize the formulation, processing, and delivery of microsphere-based products [3].

Attenuated total reflectance Fourier Transform Infrared Spectroscopy

Fourier Transform Infrared Spectroscopy (FT-IR) is a powerful analytical technique used to characterize the chemical composition and molecular structure of materials. When applied to microspheres, FT-IR helps in assessing the stability and deterioration of the carrier device polymer matrix. Attenuated Total Reflectance (ATR) is a specific FT-IR technique used to analyze the surface properties of microspheres. FT-IR spectroscopy is used to monitor the deterioration of the polymer matrix in microspheres. The technique can detect changes in the chemical structure of the polymer, which may occur due to environmental factors, manufacturing processes, or aging [141].

Microspheres are prepared for FT-IR analysis by placing them in a suitable sample holder, with solid microspheres either being ground into a fine powder or analyzed as-is if compatible with the equipment. The FT-IR spectrometer then captures the infrared spectrum, which shows the absorbance of infrared light by the microspheres as a function of wavelength. The resulting spectrum is analyzed for changes in characteristic absorption peaks, with deterioration indicated by shifts in peak positions, changes in peak intensity, or the appearance of new peaks, signaling chemical degradation or alterations in the polymer matrix [142].

Zeta Potential

The electrical charge on the surface of suspended particles is measured by eta potential, which offers information about the stability of colloidal systems like microsheres. The electrostatic repulsion between particles is determined by the zeta potential; a greater absolute zeta potential (positive or negative) suggests stronger repulsion, which lowers aggregation and increases colloidal stability. Electrophoretic light scattering (ELS), which determines the mobility of particles

under an applied electric field, is commonly used to quantify zeta potential [143]. Zetasizers and other similar instruments are frequently employed for this purpose. (Ex. Laser Doppler Electrophoresis, Microelectrophoresis, Streaming Potential Measurement, Acoustic Electrophoresis, Dynamic Light Scattering (DLS) with Zeta Potential Analysis, Capillary Electrophoresis.

In vitro Release Study

According to the monograph, the disintegrating media is the liquid present at the ingestion site. To consider the in vitro discharge profile, a pivoting bin or an oar form disintegration contraption are used in standard IP/BP/USP disintegration mechanical assemblies.

In Vitro Dissolution Test for Multiparticulates

Several techniques are suggested by USP 42-NF 37 for the dissolution of multiarticulate preparations [144]. The dosage form's performance in the medium and the volume are taken into consideration when selecting the apparatus and volume. When sink conditions are met, a flow-through cell (USP Apparatus 4) can be used for multiple dosage forms (beaded products), or a reciprocating cylinder (USP Apparatus 3) can be used for non-disintegrating coated beads. Ionic polymers behave differently at different pH values, which affects how drugs release. Different pharmacopoeia techniques can be used to ascertain whether drug release is independent of or dependent upon the ionic strength of the medium. These techniques use pH-1.2–7.5 media, such as phosphate buffers, acetate buffers, or hydrochloric acid. Depending on the route of administration or the behavior of the polymer in the medium (like swelling or ionization at the isoelectric point, IEP), enzymes may be added. Surfactants can also be used, although usually at concentrations higher than their critical micelle concentration (CMC). As a noncompendial apparatus, a mini-paddle-equipped small-volume apparatus, dialysis tubes, or rotating bottle can be used [27, 145].

Dissolution Test for Inhaled Particles

Particles that are inhaled can deposit in the tracheobronchial tract according to their size, density, and form. As the site of drug absorption, the mucus layer has a limited volume and renews in a different way than the gastrointestinal tract. As such, dissolution techniques that are customized for this particular environment are required, which deviates from conventional techniques. One such apparatus that can be modified for this use is USP Apparatus 2, or the rotating paddle apparatus. With this altered configuration, the medication is put into a specific cassette, and tiny paddles are used to stir the 150 mL medium during the dissolution process. An additional practical method for researching drug dissolution is the dialysis bag method. Using this technique, the medication is put inside a dialysis bag that is semipermeable, and it is submerged in the solution. Between the two liquid phases of the dialysis membrane, the medication diffuses. The air-liquid interface that occurs in the respiratory tract is not replicated by this method, despite sink conditions and static conditions being maintained to make the study easier [146].

In Vitro Performance of Intramuscular Injections

In order to accommodate the small interstitial volume characteristic of this formulation, the in vitro dissolution study of the long-acting intramuscular injection Risperdal® Consta® made use of a flow-through cell apparatus with 1 mm diameter glass beads and a pump. This configuration evaluated the dissolution process in an environment similar to that of the drug in vivo. The dissolution behavior of the polymer in the accelerated test, which was carried out at a temperature higher than its glass transition temperature, showed a strong correlation with previously published in vivo data. The drug powder is poured onto a semipermeable membrane in the Transwell® system and the modified Franz diffusion cell. The membrane is positioned above the acceptor medium and the drug diffuses through it during the test. The in vivo performance can be simulated in vitro under sink conditions, which include small volume, agitation, and regular sample withdrawals [147].

Physical Stability

The ability of a microsphere to preserve its encapsulated content and structural integrity over time is referred to as physical stability. Particle size, bilayer composition, zeta potential, and environmental conditions (pH, temperature) all have an impact on physical stability. Preserving stability is essential to the longevity of medication formulations in microspheres. Microsphere stability is frequently assessed through long-term storage studies. In these investigations, microspheres are stored at different temperatures and analyzed using TEM/SEM for morphological changes, HPLC or UV-Vis spectroscopy for drug leakage, and DLS for size changes [114].

Biocompatibility

The term "biocompatibility" describes microsphere capacity to operate in a biological setting without producing unfavourable outcomes including toxicity, tissue damage, or immunological reactions. This characteristic is essential to guaranteeing that patients can receive microsphere preparations safely. Because micros employ non-ionic surfactants, which are less hazardous than cationic or anionic surfactants, they are typically thought to have superior biocompatibility. For systemic delivery applications and long-term therapy, maintaining biocompatibility is essential. Biocompatibility is measured using in vivo investigations to examine the effect on living tissues and in vitro cytotoxicity assays (e.g., MTT or LDH assays) [148].

Toxicological Effects

The term "cytotoxicity" refers to microsphere' ability to halt or kill cell growth, which is essential for both safety and effectiveness. Although microsphere are normally designed to have as little negative impact on healthy cells as possible, they can be engineered to cause cytotoxicity in particular diseased cells, such as cancer cells. The drug's cytotoxicity is dependent on the target cell type, the drug contained in the microsphere, and the makeup of the microsphere. Cytotoxicity is determined using cell viability assays, such as MTT, trypan blue exclusion, or flow cytometry-based apoptosis tests. These assays determine the percentage of live cells that remain after being exposed to the microsphere formulation [149].

Applications

Ophthalmic Drug Delivery

Polymers are perfect for creating ocular drug delivery systems because they have favourable biological features like bioadhesion, permeability enhancement, and distinct physico-chemical characteristics. Because hydrogels are elastic, they are more palatable than solid or semisolid formulations, such as ointments or suspensions. In particular, ophthalmic chitosan gels improve adherence to mucin, which coats the corneal surface and conjunctiva. This prolongs the drug's precorneal residence time and decreases lachrymal flow-mediated elimination [150]. Better drug penetration, more focused effects, and possibly lower drug dosages are the outcomes of this. On the other hand, polymer-based colloidal systems work well as drug carriers across the mucosa. Drugs such as indomethacin, for instance, can be more easily transported to the inner eye through chitosan-coated colloidal systems, whereas cyclosporine-containing chitosan microparticulates can build up in the corneal and conjunctival epithelia. Furthermore, topical delivery of medications like acyclovir to the eye appears to be a promising application for micro particulate carriers like microspheres [151].

Gene Delivery

A variety of techniques are used in gene delivery systems, such as cationic liposomes, polycation complexes, viral vectors, and microencapsulated systems. Although viral vectors are very effective and can target a wide variety of cells, they can also trigger immunological reactions and possibly cause cancer. Gene therapy non-viral delivery systems are being investigated more and more to overcome these constraints. Non-viral delivery systems come with a number of benefits, including ease of preparation, targeted delivery to particular cells or tissues, reduced immunological response, support for plasmid sizes without restrictions, and large-scale, repeatable production [152]. Particularly polymers have demonstrated promise as DNA carriers in applications involving gene delivery. They can effectively navigate the gastrointestinal (GI) tract, which makes them suitable for oral gene delivery due to their adhesive and transport properties. For example, MacLaughlin et al. showed that chitosan and depolymerized chitosan oligomers could be used to effectively deliver plasmid DNA containing a luciferase reporter gene and a cytomegalovirus promoter sequence in vivo. The luciferase gene was able to be expressed in the intestinal tract thanks to this delivery method, demonstrating the potential of chitosan-based polymers for oral gene delivery [153].

Intratumoral and Local Drug Delivery

Intratumoral and local drug delivery strategies have recently emerged as promising approaches in cancer therapy. To deliver paclitaxel directly to the tumor site at therapeutically relevant concentrations, polymer films were developed. These films, which are translucent and flexible, were able to load paclitaxel at a concentration of 31% (w/w). The films were produced using a casting method, which achieved high loading efficiencies while maintaining the chemical integrity of paclitaxel throughout the preparation process [154].

Using diazepam, the use of polymer films for oral drug delivery in rabbits was studied [155]. According to the study, a film with a drug-to-polymer ratio of 1:0.5 could function as an efficient dosage form that is similar to formulations found in commercial tablets. The polymer's potential as a replacement for traditional pharmaceutical tablets is indicated by its capacity to form films. The polymer's unique suitability for oral drug delivery applications is further highlighted by its sensitivity to pH and the reactivity of its primary amine groups [156].

Nasal Drug Delivery

A number of polymer salts, such as chitosan lactate, chitosan aspartate, chitosan glutamate, and chitosan hydrochloride, have shown promise for the prolonged release of vancomycin hydrochloride in nasal formulations. The nasal mucosa has favorable properties for drug absorption, making it a prime location for bioadhesive drug delivery systems. Drug delivery systems based on polymers, such as gels, liposomes, and microspheres, have good bioadhesive qualities and expand when they come into contact with the nasal mucosa. This swelling increases the medication's bioavailability and lengthens its half-life when taken by route [157]. When diphtheria toxoid encapsulated in chitosan microparticles is administered nasally, for example, it has been demonstrated to elicit systemic and local immune responses, including increased IgG production against diphtheria toxoid. Significant serum IgG responses and similar levels of secretory IgA have been produced by these nasal formulations, frequently outperforming immune responses obtained by parenteral vaccination administration. It has been discovered that the most effective formulation for insulin delivery in sheep is nasal administration of insulin incorporated into polymer powders, which outperforms both chitosan nanoparticles and chitosan solutions in terms of nasal drug delivery efficacy [158].

Buccal Drug Delivery

Characteristics that promote improved absorption and prolonged drug release. With regard to buccal delivery systems, chitosan in particular shows promise due to its capacity to stick to mucosal surfaces and enhance absorption. For example, chitosan microspheres combined with chlorhexidine diacetate can offer prolonged drug release in the buccal cavity, enhancing the medication's antimicrobial efficacy. Additionally, because of the inherent qualities of the polymer, chitosan microparticles can have antimicrobial effects even in the absence of an active medication. For controlled drug delivery in the oral cavity, buccal bilayered devices such as layered tablets or bilaminated film that combine chitosan with medications like nifedipine and propranolol hydrochloride exhibit great promise. To further optimize drug release profiles and overall effectiveness, anionic crosslinking polymers, such as polycarbophil, sodium alginate, or gellan gum, can be added to these systems [159, 160].

Gastrointestinal Drug Delivery

Polymer-based drug delivery systems with controlled release and buoyancy can greatly improve intestinal drug delivery. Granules of polymer that have internal cavities and are formed by deacidification float when placed in neutral or acidic media. Because of its buoyancy, medications like prednisolone can be released gradually, resulting in longer-lasting therapeutic effects. Another type of gastroretentive controlled release delivery system is melatonin contained in hollow, floating microcapsules. In simulated gastric fluid, these microcapsules successfully extend the duration of drug release, reducing melatonin release rates to a range of 1.75–6.7 h. Mucoadhesive microcapsules are especially useful for keeping medications in the stomach for prolonged periods of time [161]. For example, chitosan microspheres containing medications like glipizide and metoclopramide show long retention periods often longer than 10 h. Because of this prolonged retention, the medications remain in the stomach for longer periods of time, improving their therapeutic effectiveness [162].

Vaginal Drug Delivery

Thioglycolic acid modification of primary amino groups significantly improves the mucoadhesive properties of polymers. This alteration is especially helpful for polymers that contain clotrimazole, an imidazole derivative that is used to treat genitourinary tract mycotic infections. When thiol groups are added, the polymer's mucoadhesion is significantly increased, enabling it to stick to the vaginal mucosa for about 26 times longer than when the polymer is left unaltered. Controlled drug release is ensured by this prolonged residence period, which is essential for treating mycotic infections. metronidazole and acriflavine-containing polymers used in the formulation of vaginal tablets have good adhesion qualities and acceptable drug release profiles. These characteristics, which provide consistent drug delivery and allow for extended contact with the mucosal tissue, help them be effective in treating infections [163, 164].

Transdermal Drug Delivery

When creating drug delivery systems where drug release can be precisely controlled by varying membrane thickness and cross-linking, polymers with superior film-forming qualities are essential. For example, chitosan-alginate polyelectrolyte complexes are being investigated for a range of applications, such as wound dressings, controlled release systems, and packaging. These complexes are formed in situ into beads and microspheres. Because of their biocompatibility and biodegradability, chitosan-alginate gel beads are useful delivery systems for localized medication administration. These gel beads offer prolonged release for medications such as prednisolone, which boosts therapeutic efficacy and improves local inflammation treatment results [165]. The kind of membrane used in these systems affects how quickly drugs are released. For instance, chitosan membrane and hydrogel combined with the local anaesthetic lidocaine

hydrochloride has shown to be a successful transparent drug delivery system. This configuration is appropriate for applications requiring consistent and dependable drug administration because it provides precise control over release kinetics [166].

Imagine and Diagnostic

Microspheres have shown great promise in the fields of imaging and diagnostics, especially as biosensors and contrast agents. For example, microspheres in contrast agents have been engineered to perform better in imaging modalities like MRI and ultrasound. Microspheres encapsulated in particular materials or filled with gas function as contrast agents in ultrasonic imaging, enhancing the visibility of tissue structures and blood flow. One prominent instance is the use of perfluorocarbon microspheres as contrast agents in ultrasonography, which greatly improves the imaging's resolution and diagnostic potential by making it easier to distinguish between organ boundaries and blood vessels. On the other hand, by enabling highly sensitive and specific biomolecular assays, biosensors utilizing functionalized microspheres are advancing the early detection of diseases. It is possible to coat or embed these microspheres with biochemical ligands that bind to target biomarkers selectively [167]. For instance, in assays where the presence of these antigens is detected by measuring changes in light scattering or fluorescence, microspheres functionalized with antibodies against particular cancer antigens can be used in the detection of cancer biomarkers. The application of magnetic microspheres in immunoassays to identify prostate-specific antigen (PSA) in blood samples serves as an example. Using their magnetic properties and fluorescence readouts, these microspheres coated with PSA-opposing antibodies enable highly sensitive detection, helping to facilitate an early and precise diagnosis of prostate cancer [168, 169].

Environmental Application

As pollutants and catalysts, microspheres have demonstrated great promise in environmental applications. The use of microspheres in environmental cleanup operations can significantly improve the extraction and removal of pollutants from both air and water. One such example is the use of magnetic microspheres in water treatment. These microspheres are functionalized with materials that can take up heavy metals or organic pollutants. After the contaminants have been bound to the microspheres, an external magnetic field can be applied to them to increase the process's efficiency and economy. For example, iron oxide microspheres are widely used as a practical solution to water pollution issues by removing lead and arsenic from drinking water [170, 171].

Microspheres are designed to increase chemical reactions through catalysis by providing a high surface area and customized reactivity. For instance, microspheres made from catalysts like zeolites or metal-organic frameworks (MOFs) are used in the catalytic conversion of waste plastics into useful fuels. The polymer chains in plastic waste are broken down into simpler hydrocarbons that can be refined into fuel with the help of these microspheres. Because of the microspheres' large surface area, the catalytic process is more efficient, resulting in faster and more efficient conversion. This strategy turns plastic waste into useful resources, which not only helps recycle the waste but also lessens its impact on the environment [172].

Recent Advancements of Microsphere

Superhydrophobic Surface-Assisted Preparation of Microspheres

The capacity of superhydrophobic surfaces (SHS) to reduce the water/solid contact interface has led to their widespread development. Water droplets remain almost spherical when deposited onto SHS at contact angles greater than 150°, and the low stickiness of the SHS makes it simple to remove the droplets when the substrate is tilted. Because of these special qualities, SHS is excellent for a variety of uses. The creation of supraparticle materials and microspheres is one particularly exciting application. For the fabrication of microspheres and supraparticles, SHS offers a flexible and efficient platform that offers customization and accuracy for a range of applications. The main tactics used to create microspheres and supraparticles using SHS are described in this review [173]. These tactics include droplet template evaporation methods, polymer melting procedures, and cross-linking processes. The creation of photonic devices with adjustable structures and structural colors, the use of microspheres as catalysts with novel or complementary qualities, their incorporation into medical devices for a range of uses, the stimulation of protein crystallization, and the detection of minute amounts of analytes are a few examples of these applications. The special qualities of SHS benefit every application, improving efficiency and opening doors for creative thinking in these domains. An emerging class of materials with highly tunable properties are called supraparticles [173]. The size, shape, structure, and other characteristics of supraparticles can be precisely controlled by carefully crafting the colloidal formula and choosing droplets with the right diameter. This versatility allows for

a variety of uses, including photonic crystals and catalytic processes. Because of their high surface area and uniform size, microspheres are indispensable in many fields. They work especially well as carriers for precise drug release control and high-volume encapsulation of cells. In addition, SHS-driven procedures like protein crystallization and SHS-assisted trace solute detection frequently result in the production of materials that resemble microspheres [173]. The incorporation of superhydrophobic surfaces into microsphere fabrication has led to transformative advancements in precision drug delivery and diagnostics. For instance, Demonstrated the use of superhydrophobic templates to create microspheres with dual-functional properties for cancer treatment and imaging. These microspheres were loaded with doxorubicin for chemotherapy and iron oxide nanoparticles for MRI contrast enhancement, enabling theranostic capabilities in a single system. Such innovations represent a paradigm shift from conventional drug delivery methods to multifunctional platforms tailored for personalized medicine. Superhydrophobic surfaces are now being employed in photonic devices for biosensing, as illustrated in a recent study, where microspheres enhanced light-guided diagnostics for real-time biomarker detection [173, 174].

Probiotics-Loaded Microspheres for Cosmetic Applications

Among the many health advantages of probiotic bacteria are their ability to support the intestinal microbiome and prevent antibiotic resistance. Probiotic products have long been used in the food and pharmaceutical industries, but in recent years, their use in the cosmetics industry has grown. Nevertheless, there are difficulties involved in adding live bacteria to cosmetic formulations, especially when it comes to upholding microbiological standards and guaranteeing cell viability. In order to overcome these obstacles, a recent study created a successful technique for adding live *Lactobacillus casei* to cosmetic goods. To enhance bacterial viability, a novel encapsulation technique was employed. Alginate microspheres were specifically made as carriers to protect the probiotic strain against outside influences like UV light, temperature changes, and preservatives. The study discovered that *L. casei* was well-protected during storage by alginate microspheres. These probiotic-loaded microspheres demonstrated strong active properties when added to cosmetic formulations [175]. In addition, a meticulously chosen preservative system was employed to preserve the microorganisms' viability as well as the products' overall microbiological stability. The microencapsulated probiotic strain's survival rate showed a noticeable improvement, according to the data. In contrast to a formulation containing non-immobilized probiotic cells, which showed total bacterial cell elimination, the formulation containing the encapsulated L. casei showed a viable count of 6.13 log CFU/g after 120 days of storage. The microencapsulation process offers a promising solution to the challenge of preservative incompatibility, hence prolonging the bacteria's life in skin care products. Particularly, alginate microparticles are made of L-guluronic and D-mannuronic acid molecules that are biocompatible and biodegradable and are connected by glycosidic bonds. The use of alginate microparticles in food, pharmaceutical, and environmental applications have dominated most scientific literature; there is still much to learn about their potential in cosmetic formulations. Alginate microparticles have the potential to encapsulate different active ingredients in cosmetic products. These include probiotics and other non-toxic active ingredients, as well as live strains of predatory bacteria with antimicrobial qualities, like *Bdellovibrio bacteriovorus*. Investigating the use of alginate microparticles in cosmetic formulations may result in creative ways to deliver and preserve healthy ingredients [175]. Advancements in probiotics-loaded microspheres have redefined their role from dietary supplements to targeted therapeutics. A recent study encapsulated *Lactobacillus rhamnosus* in alginate-chitosan microspheres using a double-coating method. The results demonstrated enhanced survival rates in gastric environments and controlled release in the intestine, improving efficacy in treating irritable bowel syndrome (IBS). Another study highlighted the use of symbiotic microspheres containing prebiotics and probiotics for modulating gut microbiota in patients with metabolic syndrome. These examples illustrate a paradigm shift toward precision nutrition and therapeutic interventions, bridging the gap between pharmaceuticals and nutraceuticals [176, 177].

Microspheres for Cancer Therapy

Embolization Therapy

The use of biomaterials in embolization therapy has greatly improved liver cancer treatment approaches and raised patient survival rates. In order to improve therapeutic outcomes, biomaterials have been developed as embolic agents. These agents come in a variety of forms, such as foams, particulates, liquids, and microspheres. Two of the most widely used embolic agents are microspheres, more precisely three-dimensional micrometer-sized spherical particles. Research is currently being conducted to add more functions to these microspheres in order to increase their efficacy in therapeutic embolization [178]. Several of the drawbacks of conventional cancer treatments are mitigated by their large surface area and ability to be loaded with radionuclides or other active therapeutic agents. Microspheres, both non-radioactive and radioactive, have been extensively used and studied

for localized cancer treatment. The ability of non-radioactive microspheres to deliver drugs during chemotherapy makes them especially valuable. They enhance clinical results by delivering sustained and regulated drug release to specific locations [179]. Their delivery via injection procedures is further facilitated by their advantageous flow properties. As an alternate method of treatment, radioactive microspheres are used as an embolic platform for internal radiotherapy. They enhance conventional cancer treatment modalities by providing the advantages of localized radiation therapy right to the tumor site [180].

Glioblastoma Multiforme

Approximately 80% of all malignant brain tumors are glioblastoma multiforme (GBM), the most aggressive primary brain tumor. The blood-brain barrier (BBB) presents challenges that contribute to GBM's poor therapeutic prognosis and low survival rates. One promising way to get around this problem is to inject drug-loaded microspheres locally and specifically. This technique allows for sustained, localized drug release right at the tumor site while also avoiding the blood-brain barrier. For example, by encasing vincristine (VCR) in magnetic albumin microspheres (MAMbs) and conjugating anti-CD133 monoclonal antibodies (CD133mAb) to their surface, researchers have created CD133mAb/TMAMbs, a targeted drug delivery system. The ability of this targeted delivery method to precisely bind to CD133 receptors on U251 glioblastoma cell membranes was demonstrated. As soon as it was delivered, VCR began to effectively suppress the growth of tumor cells [181, 182].

Although mitoxantrone has strong anti-glioblastoma multiforme (GBM) activity in vitro, its effectiveness is restricted by poor penetration into the central nervous system. Mitoxantrone can be encapsulated in microspheres and repeatedly injected into the tumor site to overcome this obstacle and provide long-lasting and efficient local drug concentrations. Studies in the lab have shown that this strategy greatly slows the growth of malignant glioblastoma. After one week of intratumorally injection, for example, the tumor volume in the treated subjects was about 17 mm3, which was notably less than the 76 mm^3 observed in the microsphere-receiving control group. Imatinib mesylate-loaded PLGA microspheres were the subject of additional research. When these microspheres were injected subcutaneously, the tumour volumes of murine (GL261) and human (U87-MG) glioma tumors were reduced by 88 and 79%, respectively. Furthermore, the tumor volume was 79% smaller after a single intracranial injection of the drug-loaded microspheres. These findings demonstrate the potential of microspheres as a practical and effective means of administering drugs to treat brain tumors [183, 184].

Lung Cancer

With lung cancer accounting for about 18% of cancer-related deaths globally, lung cancer continues to be the leading cause of cancer-related deaths. Targeted therapy, immunotherapy, and anti-angiogenic therapy are the main treatments for lung cancer; microspheres are important components of targeted therapy. Studies show that when injected intravenously, smaller microspheres can successfully accomplish passive targeting to the lungs. For example, it has been demonstrated that 12 µm PLGA microspheres are especially useful for passive lung targeting. Furthermore, research has shown that in mice infected with methicillin-resistant Staphylococcus aureus (MRSA), lysozyme-loaded PLGA microspheres (LyIR@MS) show promising lung-targeting abilities. Furthermore, sophoradin's therapeutic efficacy for the treatment of lung cancer may be improved by PLGA microsphere drug delivery systems (DDS). In just half an hour, sophoradin-loaded PLGA microspheres that were made by emulsification were able to reach significant drug concentrations in the lungs (220.10 mg/g). This concentration was roughly 33 times greater than the 6.77 µg/g drug levels found in the solution group [185, 186].

Prostate Cancer

A well-known example of a hormone-dependent tumor is prostate cancer, whose onset and spread are intimately correlated with sex hormones. Chemotherapy, radiotherapy, and androgen deprivation therapy are the main treatments for prostate cancer. One notable application for microspheres is in hormone deprivation therapy. The first prostate cancer treatment using microspheres was liprolide acetate (Lupron® Depot). It is an analogue of gonadotropin-releasing hormone (GnRHa) that controls hormone levels in the body by means of negative feedback mechanisms. This formulation is made by evaporating the traditional emulsion solvent method with PLGA as the carrier material. This method establishes a standard for similar drugs by enabling the continuous release of leuprolide over a month. It is currently offered in four dosage forms that call for intramuscular injections every 1, 3, 4, or 6 months [187].

However, because of the traditional emulsification technique used in their preparation, all of these dosage forms experience an initial burst of active pharmaceutical ingredient (API). This outburst may require more medication and raise the possibility of adverse reactions. With the use of IVL-DrugFluidic™ microsphere technology within a microfluidic system and micro-electromechanical systems, recent developments have brought forth a novel approach for leuprolide microsphere preparation. Studies comparing

formulations made with this new technique PLGA-based for 1 month and PLA-based for 3 months showed a number of advantages over microspheres that are sold commercially. In particular, these novel microspheres provide a more regulated and efficient therapeutic option by decreasing the initial burst release, improving testosterone inhibition, and stabilizing API concentrations in the plasma [6].

Microspheres for Diagnostic Purpose

Synthesis of Eco-Friendly Multifunctional Dextran Microbeads for Multiplexed Assays

The increasing demand for simultaneous detection of multiple analytes in a single sample has led to the development of microbead-based platforms for multiplexed assays. Traditional microbeads often rely on non-biodegradable synthetic polymers, which raise concerns regarding environmental impact and potential health risks. In response, we utilized biodegradable dextran as the primary material and successfully synthesized highly uniform magnetic dextran microspheres using the Shirasu porous glass (SPG) membrane emulsification technique. To enhance surface functionalization for ligand conjugation, we coated the dextran microbeads with dendrimers via a simple electrostatic adsorption method. We also developed a novel click chemistry coupling method to fluorescently encode the microspheres, facilitating efficient multiplexed detection. The performance of these dextran microbeads was assessed in a 3-plex cytokine analysis, revealing that they provide excellent biocompatibility, stable coding signals, low background noise, and high sensitivity, making them a promising tool for advanced multiplexed assays [188]. The use of environmentally friendly dextran microbeads in multiplex assays has been validated through specific applications. For example, a study employed dextran-based microbeads to detect multiple cytokines simultaneously in serum samples, achieving high sensitivity and specificity. Similarly, utilized dextran microbeads in environmental biosensors to detect heavy metal contamination in water. These case studies underscore the versatility of dextran microbeads in both clinical diagnostics and environmental monitoring, emphasizing their eco-friendly nature and wide-ranging applicability [188].

Hollow-Microsphere-Integrated Optofluidic Immunochip or Myocardial Infarction Biomarker Microanalysis

Development of optofluidic immunochip for the quick detection of cardiac troponin I, a critical biomarker for acute myocardial infarction (AMI), using whispering gallery mode and fibre laser enhancement. This immunochip has a unique design that uses immobilized antibodies on the inside surfaces of perforated hollow glass microspheres (HGMS) as carriers. Using very little sample is made possible by this design. The fiber laser greatly improves the immunochip's performance by increasing the limit of detection to 5 pg/mL, increasing the optical signal-to-noise ratio to 63.17 dB, and improving spectral width compression to 0.019 nm. The immunochip is very useful for early AMI risk assessment because it can yield results in as little as 15 min. Due to differences in detection principles and antibody affinities, there are some differences when compared to standard electrochemiluminescence detection methods; however, the immunochip displays a positive reference value of 0.0754 ng/mL and achieves an 88% success rate in identifying positive cases. The immunosensor, which is integrated on a polydimethylsiloxane substrate, is small and ideal for use in point-of-care devices, AMI self-screening, quick disease screening, and microanalysis of different biomarkers. There are now more opportunities for applications in these fields thanks to this development [189].

Difficulties and Challenges in the Development of Microspheres

The development of microspheres presents a range of technical and practical challenges that can significantly impact their efficacy, safety, and applicability across various fields. These difficulties arise from the complexity of microsphere fabrication, the need for precise control over their properties, and the requirements for their functional performance in diverse applications.

Stability

Because of their biological activity's sensitivity to preparation, storage, and in vivo conditions, peptides and proteins are notoriously unstable. Preventing these biomolecules from degrading in the intricate physiological environment is a significant task. Microspheres have the ability to encapsulate peptides in an effective manner that prevents denaturation and decomposition. However, the slow degradation of these microspheres can change the surrounding environment, which may have an impact on the stability of the encapsulated peptides. For example, exposed peptides and proteins may degrade when body temperature and high humidity are combined [54]. Furthermore, the stability of the encapsulated peptides may be further jeopardized by the acidic environment that arises from the slow disintegration of the microspheres. One strategy to overcome these obstacles is to make mixed microspheres that are intended to counteract the acid produced during degradation. Materials that react with and neutralize the acidic

byproducts can be added to accomplish this. As per Cai et al. (2024), adding poly(cyclohexane-1,4-diylacetonedimethylene ketal) (PCADK) to PLGA microspheres loaded with rhGH is a prominent example of this tactic. It has been demonstrated that this combination works well to lessen the adverse effects of the acidic environment created during microsphere degradation, increasing the stability of the peptides that are encapsulated [200].

Stability issues in microspheres arise due to environmental factors such as temperature, humidity, and light, which can degrade the encapsulated drug or the carrier material itself. One effective solution is lyophilization (freeze-drying), which removes moisture and stabilizes the microspheres for long-term storage without compromising their structure or function. Protective coatings, such as polymers resistant to environmental degradation, can also enhance stability by shielding the microspheres from light and oxygen exposure. Improved packaging methods, including vacuum-sealed or inert-atmosphere containers, further mitigate external factors that could compromise stability. Additionally, the incorporation of stabilizing agents, like antioxidants or cryoprotectants, into the formulation can help maintain the integrity of both the microspheres and the encapsulated drug [201].

Fabrication Challenges

One of the primary difficulties in microsphere development is achieving uniform size and shape. The fabrication methods for microspheres, such as emulsion solvent evaporation, coacervation, and electrostatic assembly, require meticulous control over numerous variables including temperature, solvent composition, and stirring rates. Small deviations in these parameters can lead to significant variability in microsphere size and shape, which in turn can affect their performance and consistency. Additionally, achieving high batch-to-batch reproducibility remains a challenge, particularly when scaling up from laboratory to industrial production. For example, the emulsion solvent evaporation technique, while widely used, often struggles with maintaining uniformity due to the complex interplay of emulsification and solvent evaporation processes [3].

Microsphere fabrication often faces challenges like maintaining uniformity, achieving scalability, and avoiding particle aggregation. Continuous manufacturing techniques, such as microfluidic systems, allow for precise control over particle size and reduce batch-to-batch variability, addressing these issues effectively. Automation of fabrication processes, including mixing, emulsification, and drying, ensures reproducibility and minimizes human error. Advanced methods like 3D printing have also been explored to fabricate microspheres with controlled morphology and drug distribution. To prevent aggregation, the use of surfactants or stabilizers during emulsification can ensure the separation of particles, resulting in more consistent microsphere batches [202].

Material Selection and Compatibility

Choosing appropriate materials for microsphere construction is another significant challenge. The materials must be compatible with the intended application, whether it be drug delivery, imaging, or diagnostics. Biodegradable polymers such as PLGA and dextran are commonly used, but they must be carefully selected and processed to ensure they meet the required specifications for drug release rates, biocompatibility, and mechanical strength. Additionally, incorporating functional groups or bioactive agents into the microsphere matrix can be challenging, requiring advanced chemical modifications and thorough testing to ensure stability and efficacy [103]. Selecting materials that are biocompatible, biodegradable, and suitable for drug encapsulation can be challenging, especially for specific applications. Natural polymers like alginate, chitosan, and gelatin provide excellent biocompatibility and biodegradability, making them ideal for pharmaceutical applications. For enhanced compatibility, hybrid materials that combine synthetic and natural polymers can balance mechanical strength with biocompatibility. Surface modification techniques, such as PEGylation, further improve material compatibility by reducing immunogenicity and enhancing circulation time in vivo. Thorough preformulation studies and computational modeling can aid in predicting the compatibility of the material with the drug and other excipients [203].

Drug Loading and Release Control

Controlling drug loading and release profiles is critical for the effectiveness of microsphere-based drug delivery systems. Achieving the desired drug encapsulation efficiency without compromising the microsphere's structural integrity is a complex task. Moreover, designing a release mechanism that provides consistent and controlled release over an extended period requires precise engineering of the microsphere's composition and structure. Variability in drug release can result in suboptimal therapeutic outcomes and increased risk of side effects. Techniques such as solvent evaporation and coacervation must be carefully optimized to balance drug loading and release rates [199]. Achieving consistent drug loading and controlled release profiles is critical for the efficacy of microspheres. Smart polymers, such as pH-sensitive or temperature-sensitive materials, allow for responsive drug release tailored to specific environments. Layer-by-layer encapsulation techniques can improve drug loading efficiency and offer multi-phase release profiles. To prevent burst release, strategies like cross-linking the polymer matrix or using core-shell structures have been

employed. For hydrophilic drugs, double emulsion methods can enhance encapsulation efficiency, while for hydrophobic drugs, solvent evaporation techniques with optimized polymer-drug ratios can provide better control [204].

Surface Functionalization

Surface modification of microspheres is essential for applications like targeted drug delivery and biomarker detection. However, functionalizing the surface to attach ligands, antibodies, or other active molecules can be technically demanding. The process must ensure that the functional agents are securely bound and retain their biological activity. Furthermore, achieving uniform functionalization across a large number of microspheres while avoiding issues like aggregation or loss of functional activity poses a significant challenge [205]. Surface functionalization is essential for improving the targeting and interaction of microspheres with specific tissues or cells. Ligand conjugation, such as attaching antibodies, peptides, or small molecules, enhances the specificity of microspheres for their target. Techniques like click chemistry and carbodiimide coupling have proven effective for precise functionalization. Nanocoatings, including polyethylene glycol (PEG) or polydopamine, can enhance biocompatibility and reduce opsonization, thereby prolonging circulation time. Incorporating fluorescent or magnetic nanoparticles during functionalization also adds theranostic capabilities, enabling simultaneous imaging and therapy [206].

Reproducibility

Reproducibility in microsphere preparation is a critical challenge that can significantly impact the reliability and effectiveness of microsphere-based applications, including drug delivery, diagnostics, and imaging. Achieving consistent reproducibility involves several intricate factors, each of which must be carefully controlled to ensure uniformity across different batches of microspheres. One of the primary issues affecting reproducibility is the sensitivity of the microsphere fabrication process to slight variations in parameters such as temperature, solvent composition, and stirring rates. For example, in techniques like emulsion solvent evaporation or coacervation, the precise control of these variables is crucial for producing microspheres with consistent size and shape. Even minor deviations can lead to significant differences in the microsphere's characteristics, including size distribution, surface morphology, and encapsulation efficiency [27, 207]. This variability can impact the performance of the microspheres, leading to inconsistent drug release rates or diagnostic signal intensities. The choice and preparation of materials play a vital role in reproducibility. The use of polymeric materials, such as PLGA or poly(lactic acid), requires stringent quality control to ensure batch-to-batch consistency. Variations in the polymer's molecular weight, degradation rate, or functional group density can affect the microsphere's properties and performance. Furthermore, the incorporation of active agents or functional groups into the microsphere matrix must be precisely managed to avoid inconsistencies in drug loading or surface functionalization [27, 208].

The scaling-up process from laboratory to industrial production also introduces challenges to reproducibility. While small-scale experiments might yield consistent results, scaling up can reveal issues related to process uniformity and material handling. Factors such as equipment differences, mixing efficiency, and solvent evaporation rates can vary on a larger scale, potentially leading to discrepancies in microsphere quality. Maintaining reproducibility across different production scales requires meticulous process optimization and rigorous quality control measures. To address these reproducibility challenges, researchers and manufacturers employ a variety of strategies. These include rigorous standardization of fabrication protocols, thorough characterization of materials and microspheres, and implementation of advanced process controls. Techniques such as real-time monitoring of the fabrication process and the use of automated systems can help minimize human error and ensure consistent results. Additionally, comprehensive testing and validation of each batch of microspheres are essential to confirm their quality and performance before use [209]. Reproducibility in microsphere production is critical for clinical translation and industrial scalability. Standardization of fabrication protocols, including controlled temperature, stirring speed, and solvent evaporation rates, ensures consistency across batches. The use of automated systems for emulsification, cross-linking, and drying minimizes variability caused by manual intervention. Continuous manufacturing processes, such as microfluidics, offer unparalleled control over particle size and distribution, significantly improving reproducibility. Quality control measures, such as real-time monitoring with spectroscopic or imaging techniques, further ensure batch-to-batch consistency and adherence to desired specifications [202].

Microspheres: A Novel Approach for Targeted Nose-to-Brain Drug Delivery

An inventive method that gets over the blood–brain barrier (BBB), a significant barrier to getting therapeutic drugs to the central nervous system (CNS), is the nose-to-brain drug delivery system. This approach allows medications to effectively reach the brain without systemic exposure by taking use of the nasal cavity's direct link to the brain via the olfactory and trigeminal pathways. In neurological conditions

where traditional medication delivery techniques frequently fail, such as Alzheimer's disease, Parkinson's disease, epilepsy, and brain tumours, this route is especially helpful. Clinical benefits of the intranasal route include increased patient acceptability because it is a non-invasive delivery approach, the ability to administer medication in the event of severe nausea or vomiting, and a more transparent record of side effects [210].

Due to their unique ability to encapsulate medications, protect them from degradation, and enable controlled release, microspheres have garnered significant interest in the nose-to-brain drug delivery system. This system leverages the nasal cavity's direct connection to the brain via the olfactory and trigeminal nerves, bypassing the restrictive blood-brain barrier that limits drug delivery to the central nervous system (CNS). Microspheres, with their small size and customizable surface properties, enhance drug absorption through the nasal mucosa and ensure precise targeting to the brain. Surface modifications, such as adding ligands or functional groups, further improve binding efficiency and targeting, making microspheres a highly effective carrier for CNS therapies. This non-invasive delivery route offers several advantages over conventional methods, including a rapid onset of action due to shorter transit times, reduced systemic side effects, and improved patient compliance, especially for chronic neurological conditions. Moreover, microspheres can accommodate a wide range of therapeutic agents, from small molecules to biologics, such as peptides, proteins, and nucleic acids. Their versatility also enables dual or multi-drug delivery, allowing for combination therapies that address complex disease mechanisms [211]. Recent advances in materials science, including biodegradable polymers like PLGA and natural materials such as chitosan, have further improved the biocompatibility, safety, and efficiency of microsphere-based delivery systems. Emerging research explores their use in treating neurodegenerative disorders like Alzheimer's and Parkinson's disease, brain tumors, and epilepsy, demonstrating their potential to transform CNS therapies. With the integration of imaging tracers or theranostic agents, microspheres also support diagnostics and therapy simultaneously, pushing the boundaries of personalized medicine and enhancing outcomes for CNS disorders [212].

Nose-to-Brain Drug Transmission Mechanism

Several pathways have been identified for medication delivery via the nose, offering distinct mechanisms to reach the brain. One approach is the systemic pathway, where medications are absorbed into the bloodstream through the nasal mucosa's vascular network. These drugs then cross the blood–brain barrier (BBB) to access the central nervous system. While effective for certain drugs, this method depends on the drug's ability to traverse the BBB. Another significant route is the olfactory pathway, which bypasses the BBB entirely. Drugs delivered through this pathway penetrate the olfactory epithelium and travel to the olfactory bulb, eventually reaching deeper brain tissues or cerebrospinal fluid (CSF). This direct access to the CNS ensures efficient and targeted drug delivery. Within the olfactory pathway, two distinct neuronal mechanisms facilitate drug transport to the brain. In the intracellular pathway, drugs are internalized by neurons via endocytosis and subsequently transported along axonal projections to the central nervous system. Alternatively, in the extracellular pathway, drugs travel through the perineural spaces surrounding the olfactory neurons, bypassing cellular transport to reach CNS structures. These complementary mechanisms leverage the nasal cavity's unique anatomical features, offering efficient and innovative solutions for delivering therapeutics directly to the brain, particularly for treating neurological disorders. Through processes like endocytosis, olfactory neurones in the olfactory epithelium take up the chemicals in the intraneuronal cascade before axonically entering the olfactory bulb [213]. Drugs administered intranasally first traverse the gaps between the olfactory nerves in the olfactory epithelium before proceeding to the olfactory bulb via the extraneuronal pathway as shown in Figure 13. After entering the olfactory bulb, the chemicals can diffuse into other parts of the brain, possibly with the help of a "perivascular pump" that is driven by artery pulsation. There is a difference between drug penetration through the nasal mucosa and drug transmission via the olfactory or trigeminal nerve systems. There is a difference between drug penetration through the nasal mucosa and drug transmission via the olfactory or trigeminal nerve systems.

The medication enters the systemic circulation through the nasal mucosa, which is highly vascularized, after being administered by the nasal cavity. It may or may not cross the blood-brain barrier and reach the brain. However, the delivery of medications into the brain is largely dependent on the olfactory epithelium, olfactory cortex, or trigeminal nerves [213].

Nose-to-Brain Targeted Microspheres for the Treatment of Alzheimer's Disease

Rivastigmine Microspheres The nose-to-brain delivery system of rivastigmine via mucoadhesive microspheres offers a promising solution to the challenges of Alzheimer's disease treatment. Alzheimer's is a progressive neurodegenerative disorder that affects cognitive function, and the blood-brain barrier (BBB) limits the efficacy of many conventional drugs. Rivastigmine, a cholinesterase inhibitor used to increase acetylcholine levels, is hindered by its inability to cross the BBB and its first-pass metabolism when taken

orally. Using mucoadhesive microspheres, rivastigmine can be administered nasally, bypassing the BBB and reaching the brain through the olfactory and trigeminal nerve pathways. These microspheres, made from biodegradable polymers like ethylcellulose and chitosan, adhere to the nasal mucosa, allowing for prolonged retention and sustained drug release, thus maintaining therapeutic drug levels over a longer period [214]. Lectin functionalization further enhances bioadhesion and drug uptake into the brain. This delivery method reduces systemic side effects, such as gastrointestinal issues and liver toxicity, commonly seen with oral or intravenous forms of the drug. By avoiding hepatic metabolism and achieving higher bioavailability in the brain, rivastigmine's effectiveness is significantly improved. The controlled release minimizes fluctuations in drug levels, ensuring a more stable therapeutic response. Rigorous optimization techniques have ensured that the microspheres offer the desired particle size, entrapment efficiency, and release kinetics. Preclinical studies, including behavioral and biochemical tests, have shown that lectin-functionalized microspheres result in better memory retention and cognitive function compared to plain rivastigmine solutions, demonstrating superior therapeutic effects. This system not only overcomes the limitations of traditional drug delivery but also represents an innovative, non-invasive approach to managing Alzheimer's disease with fewer side effects, improving patient compliance [215].

Donepezil Hydrochloride Microsphere The development of donepezil hydrochloride-loaded gellan gum-based mucoadhesive microspheres for intranasal delivery has shown significant potential in the treatment of Alzheimer's disease. Using the spray-drying technique, these microspheres were formulated with optimal particle size, high drug entrapment efficiency, and prolonged nasal residence time, all of which contribute to enhanced drug absorption through the nasal mucosa. The mucoadhesive property of gellan gum helps to maintain the microspheres' contact with the nasal lining, facilitating direct transport of the drug to the brain through the olfactory and trigeminal nerve pathways, thus bypassing the blood-brain barrier. In vitro and ex vivo studies demonstrated consistent and controlled drug release, while permeation studies showed efficient drug diffusion across the nasal mucosa. Histological studies confirmed that the formulation did not cause any damage to nasal tissues, ensuring its safety for intranasal use. Stability studies revealed minimal changes in particle size, swelling, and mucoadhesion, indicating the long-term stability of the microspheres. This approach offers a non-invasive, targeted, and effective drug delivery system for Alzheimer's, reducing systemic side effects and enhancing patient compliance. The results suggest that gellan gum-based nasal microspheres can be a promising therapeutic strategy for Alzheimer's disease, providing sustained drug release and direct brain delivery [216].

Sumatriptan Succinate Microsphere The development of hydroxypropyl methylcellulose (HPMC)-based mucoadhesive microspheres for the nasal delivery of Sumatriptan Succinate demonstrates significant potential for improving migraine management. The spray-drying technique used in this study produced microspheres with optimal particle size, high incorporation efficiency, and sustained drug release. The mucoadhesive properties of HPMC ensured prolonged

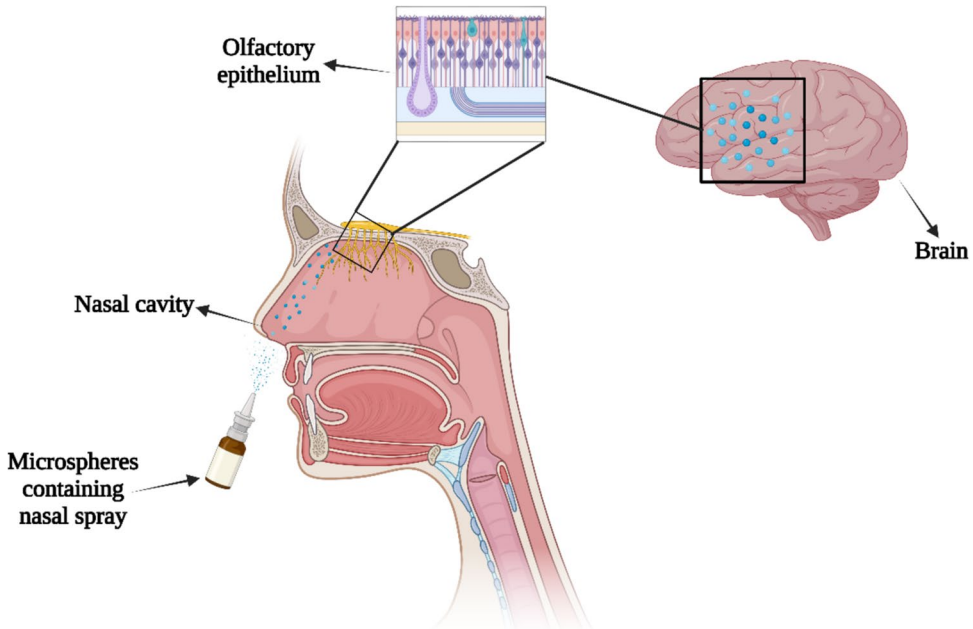


Fig. 13 Nose to brain drug transmission mechanism

residence time on the nasal mucosa, facilitating enhanced drug absorption via the olfactory and trigeminal pathways. This approach bypasses the gastrointestinal and hepatic first-pass metabolism, overcoming the limitations of low oral bioavailability and reducing the side effects associated with systemic drug delivery. Histological studies confirmed the safety of the microspheres, with no adverse effects observed on nasal tissues. Stability studies further validated the robustness of the formulation under stress conditions. These findings highlight the potential of nasal mucoadhesive microspheres as a patient-friendly, non-invasive alternative for effective and rapid migraine relief, offering significant improvements over conventional delivery methods [217].

Nose-to-Brain Targeted Microspheres for the Treatment of Migraine

Flunarizine Microspheres The development of flunarizine-loaded PLGA-based microspheres offers a promising solution for the prophylaxis of migraine. By utilizing a solvent evaporation method and an optimized PLGA 502H and PLGA 502 polymer ratio, the microspheres achieved high encapsulation efficiency (91.9 ± 1.74%), spherical morphology, and controlled release kinetics. The system demonstrated a biphasic release profile, with an initial burst followed by sustained zero-order release for 14 days, achieving a constant rate of 168.98 µg/day/10 mg microspheres. Gamma-irradiation sterilization preserved the physical and chemical properties of the microspheres, ensuring safety and efficacy. Stability studies confirmed long-term storage capabilities at 5°C for up to 18 months without altering release behavior. Cell viability and reactive oxygen species (ROS) studies showed no significant toxicity, validating the system's biocompatibility. This controlled release delivery system reduces dosing frequency, improves treatment adherence, and minimizes side effects, addressing key challenges in migraine management and enhancing patient quality of life [218].

Sumatriptan Microsphere The nasal delivery of sumatriptan represents a promising advancement for the treatment of acute migraines, offering several advantages over traditional oral and injectable routes. The use of innovative delivery systems, such as mucoadhesive nanoparticles and nasal powders, enhances drug bioavailability by bypassing the gastrointestinal tract and hepatic first-pass metabolism. This results in faster drug absorption and a quicker onset of action, providing relief within minutes. Moreover, nasal formulations are non-invasive, user-friendly, and particularly suitable for patients experiencing nausea and vomiting during migraine attacks. Studies have demonstrated that sumatriptan nasal formulations achieve better patient compliance and reduced systemic side effects compared to oral alternatives. Challenges such as potential nasal irritation and variability in absorption due to mucociliary clearance remain. Overall, intranasal sumatriptan delivery systems hold significant potential as effective, safe, and convenient options for managing migraine symptoms, improving both therapeutic outcomes and patient quality of life [219].

Rizatriptan Microsphere The development of rizatriptan benzoate-loaded natural polysaccharide-based microspheres for nasal delivery offers a promising solution for migraine treatment. Utilizing fenugreek seed polysaccharides as a mucoadhesive polymer, the microspheres were prepared via the emulsion crosslinking method, achieving high encapsulation efficiency (up to 79.22%) and optimal particle size (40.82–62.48 µm). The formulation exhibited controlled drug release, following a zero-order kinetic model, with sustained release over an extended period, effectively addressing the short half-life of rizatriptan. Stability studies confirmed that the microspheres remained stable under various conditions without significant changes in entrapment efficiency or physical properties. The use of the nasal route bypasses hepatic first-pass metabolism, improving bioavailability and ensuring rapid onset of action, crucial for managing acute migraines. Histological studies confirmed the biocompatibility of the formulation, making it a safe and effective alternative to conventional oral or injectable treatments. This innovative system enhances patient compliance and therapeutic efficacy, offering a novel, non-invasive approach for migraine management [219].

Conclusion

Microspheres have emerged as versatile and efficient drug delivery systems, offering significant advancements in controlled and targeted drug release. Their ability to encapsulate both hydrophilic and hydrophobic drugs, combined with the use of natural and synthetic polymers, has revolutionized therapeutic approaches across various fields. This review highlighted diverse preparation methods such as single and double emulsion techniques, polymerization, spray drying, and phase separation coacervation, emphasizing their advantages, limitations, and applications. Types of microspheres, including bioadhesive, magnetic, floating, radioactive, and polymeric, have demonstrated their potential in addressing critical medical challenges, including cancer therapy, vaccine delivery, and diagnostics. Recent advancements, such as the incorporation of superhydrophobic surfaces and microfluidic technology, have enhanced the precision and scalability of microsphere fabrication. Despite these innovations,

challenges remain, including low encapsulation efficiency, reproducibility, and scalability, underscoring the need for further research. Future directions should focus on developing eco-friendly preparation techniques, optimizing drug loading, and leveraging interdisciplinary approaches to overcome biological and technical barriers. Microspheres continue to hold immense promise in advancing personalized medicine and improving therapeutic outcomes, making them indispensable tools in modern pharmaceutical innovation.

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